rbhagava_Homework4

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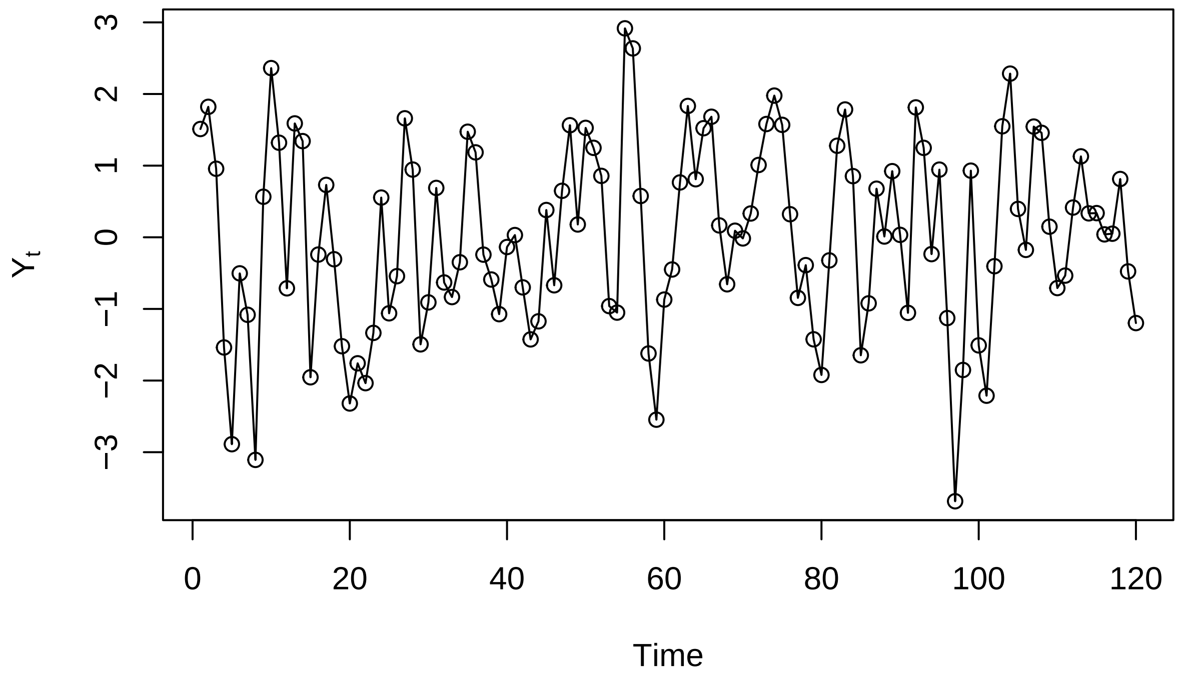
Chapter 4

```r
# Exhibit 4.2
library(TSA)
```

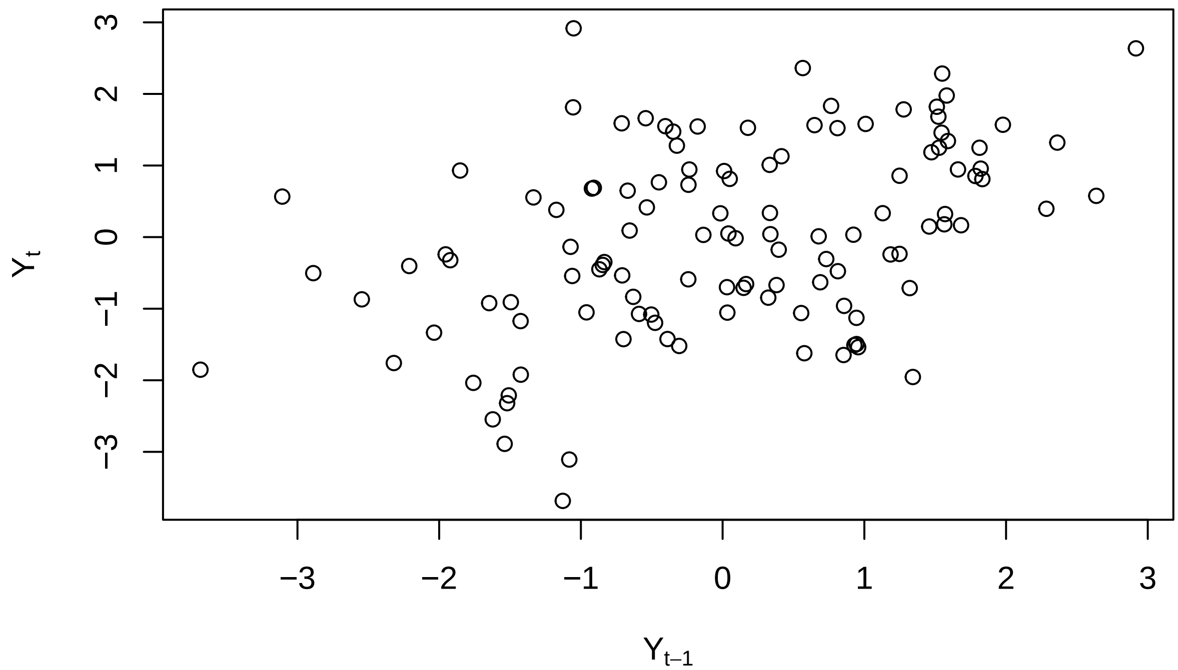
```
##
## Attaching package: 'TSA'

## The following objects are masked from 'package:stats':
##
##     acf, arima

## The following object is masked from 'package:utils':
##
##     tar
```

```r
data(ma1.2.s)
plot(ma1.2.s,ylab=expression(Y[t]),type='o')
```

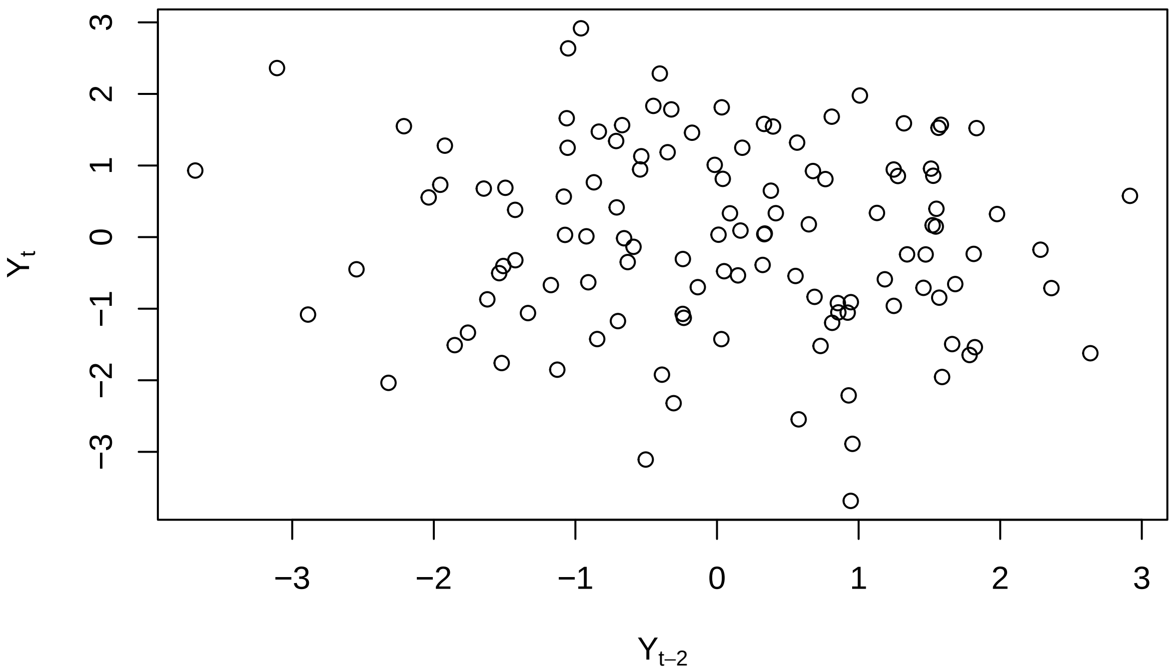



```
set.seed(12345)
y=arima.sim(model=list(ma=-c(-0.9)),n=100)
```

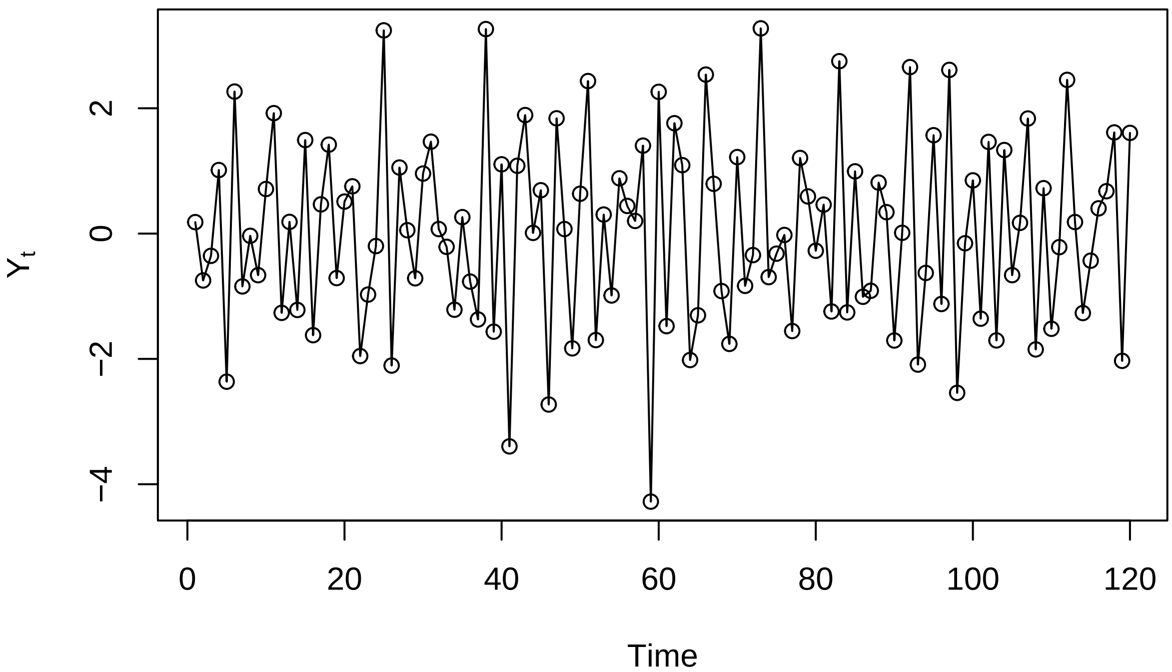
```
# Exhibit 4.3
plot(y=ma1.2.s,x=zlag(ma1.2.s),ylab=expression(Y[t]),xlab=expression(Y[t-1]),type='p')
```




```
# Exhibit 4.4
plot(y=ma1.2.s,x=zlag(ma1.2.s,2),ylab=expression(Y[t]),xlab=expression(Y[t-2]),type='p')
```

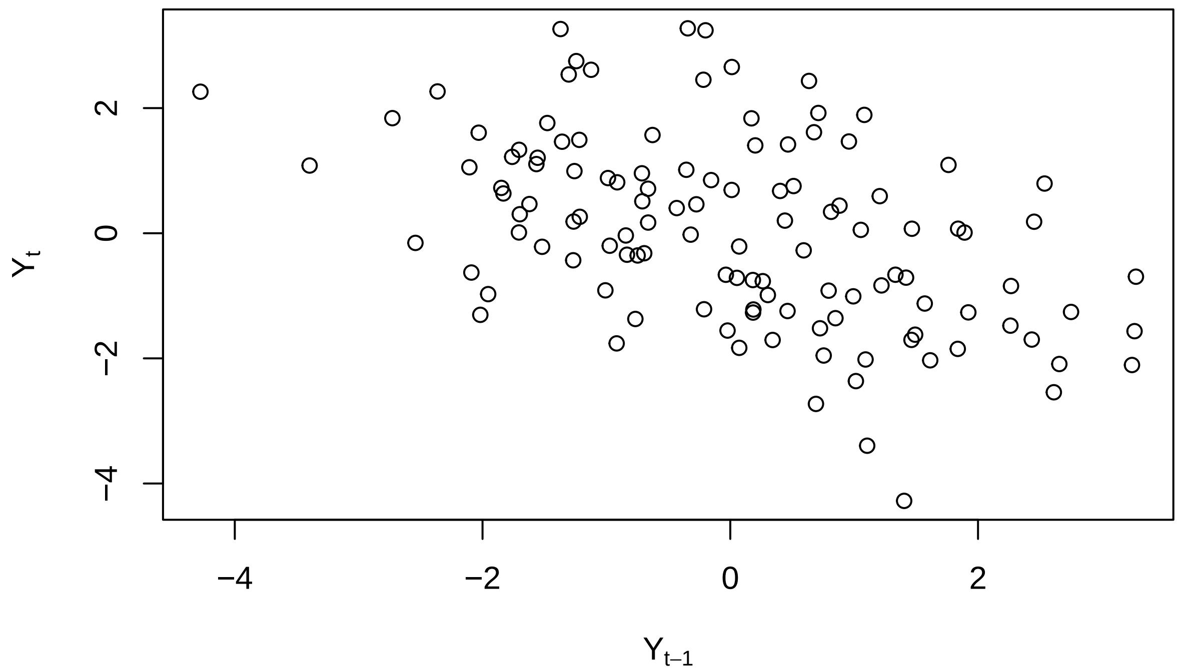



```
# Exhibit 4.5
data(ma1.1.s)
plot(ma1.1.s,ylab=expression(Y[t]),type='o')
```

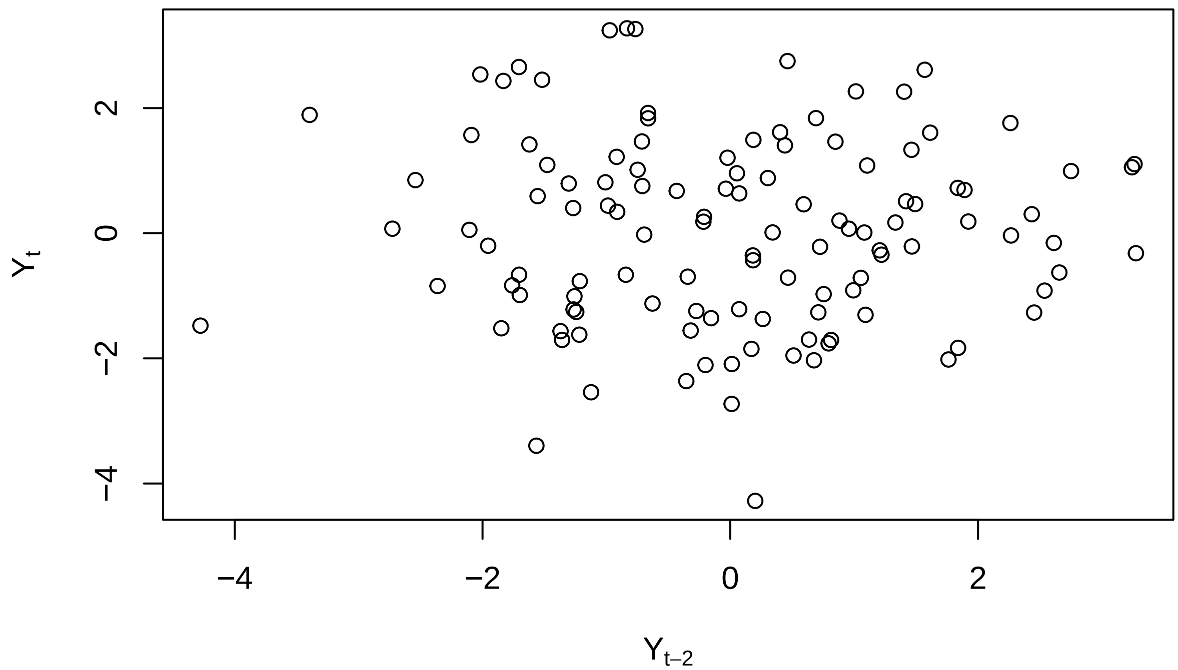



```
y=arima.sim(model=list(MA=-c(0.9)),n=100)
```

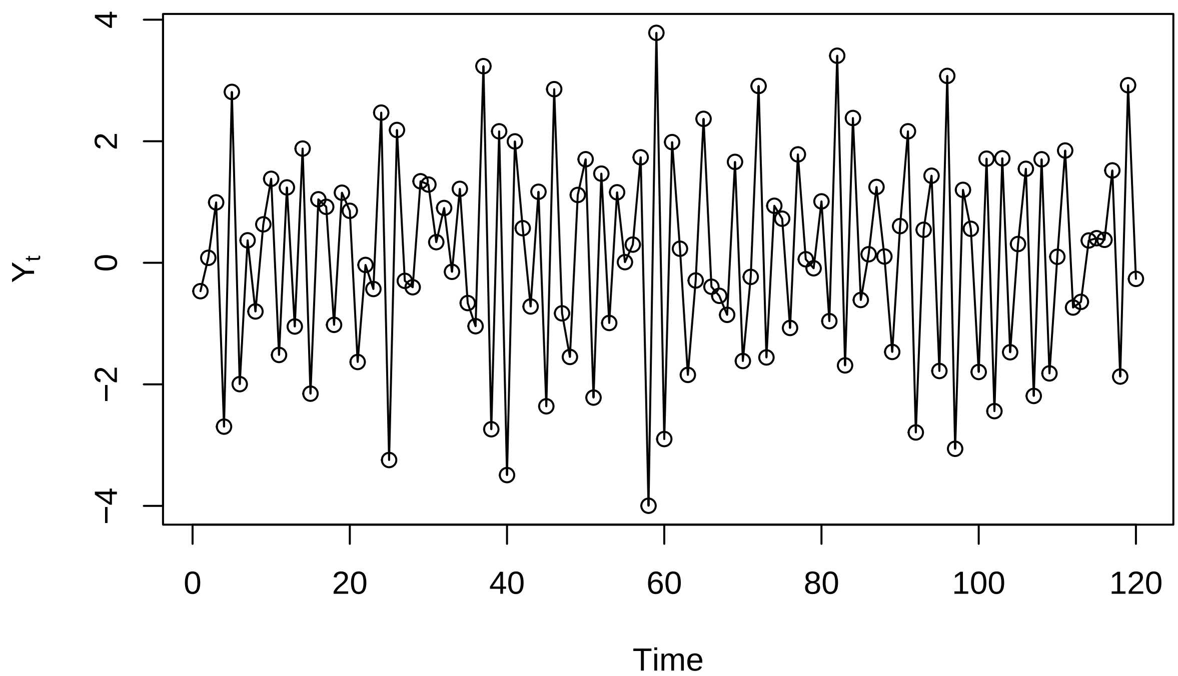
```
# Exhibit 4.6
plot(y=ma1.1.s,x=zlag(ma1.1.s),ylab=expression(Y[t]),xlab=expression(Y[t-1]),type='p')
```



```r
# Exhibit 4.7
plot(y=ma1.1.s,x=zlag(ma1.1.s,2),ylab=expression(Y[t]),xlab=expression(Y[t-2]),type='p')
```

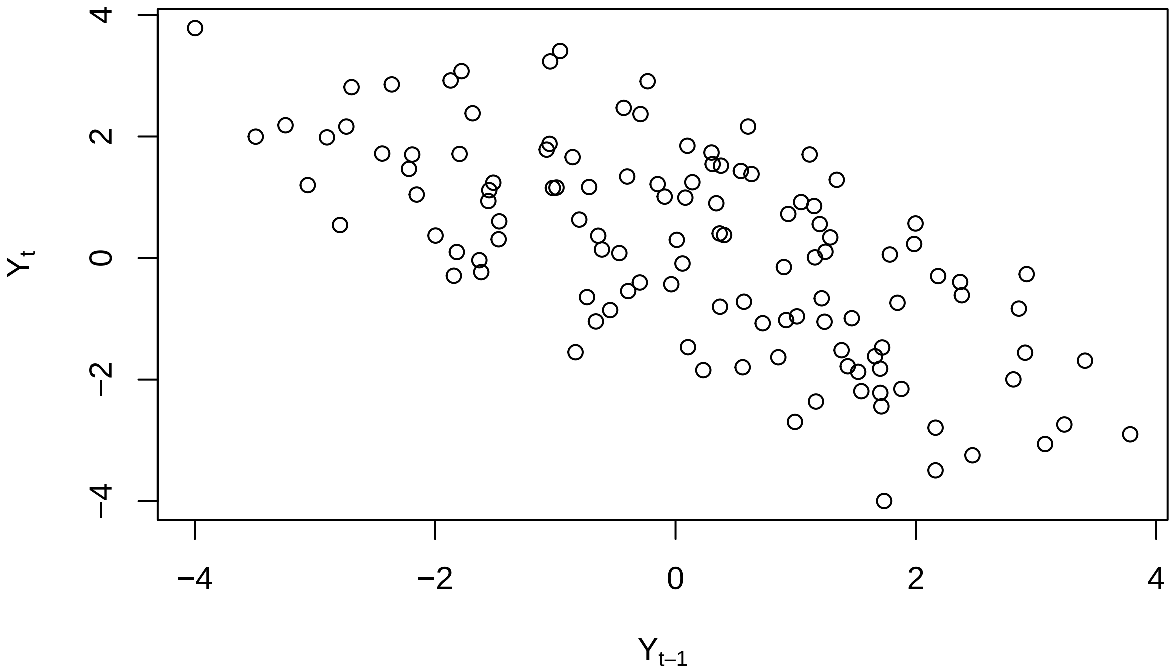



```r
# Exhibit 4.8
data(ma2.s)
plot(ma2.s,ylab=expression(Y[t]),type='o')
```

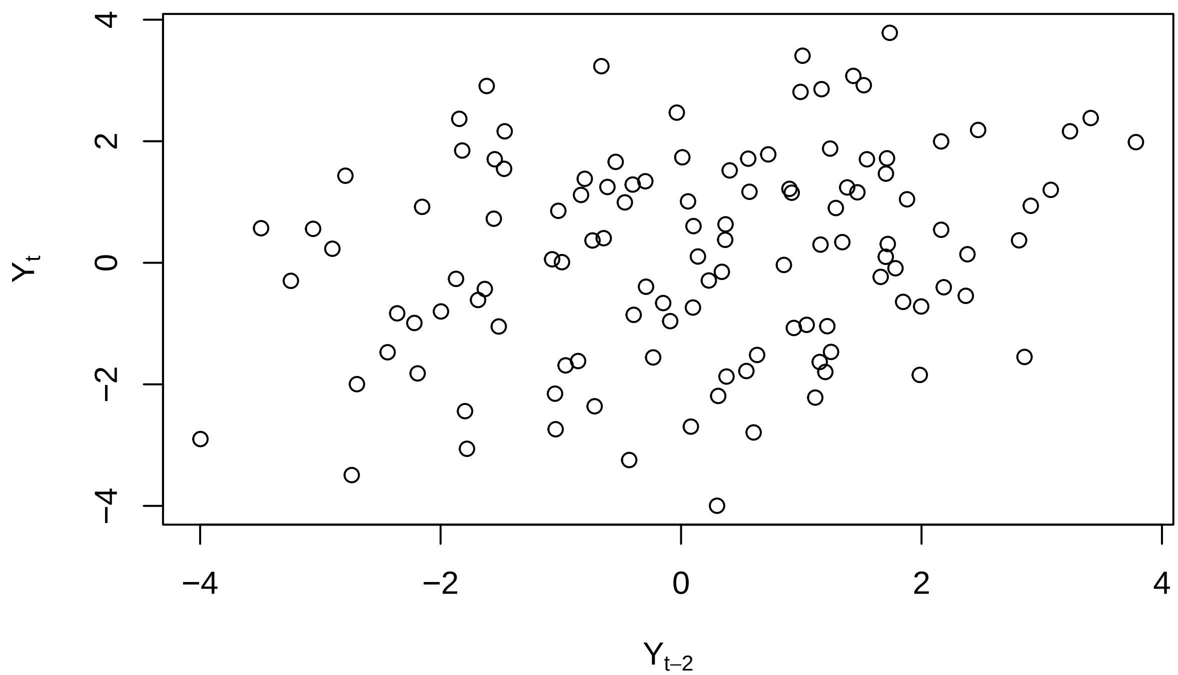



```r
y=arima.sim(model=list(ma=-c(1, -0.6)),n=100)
```

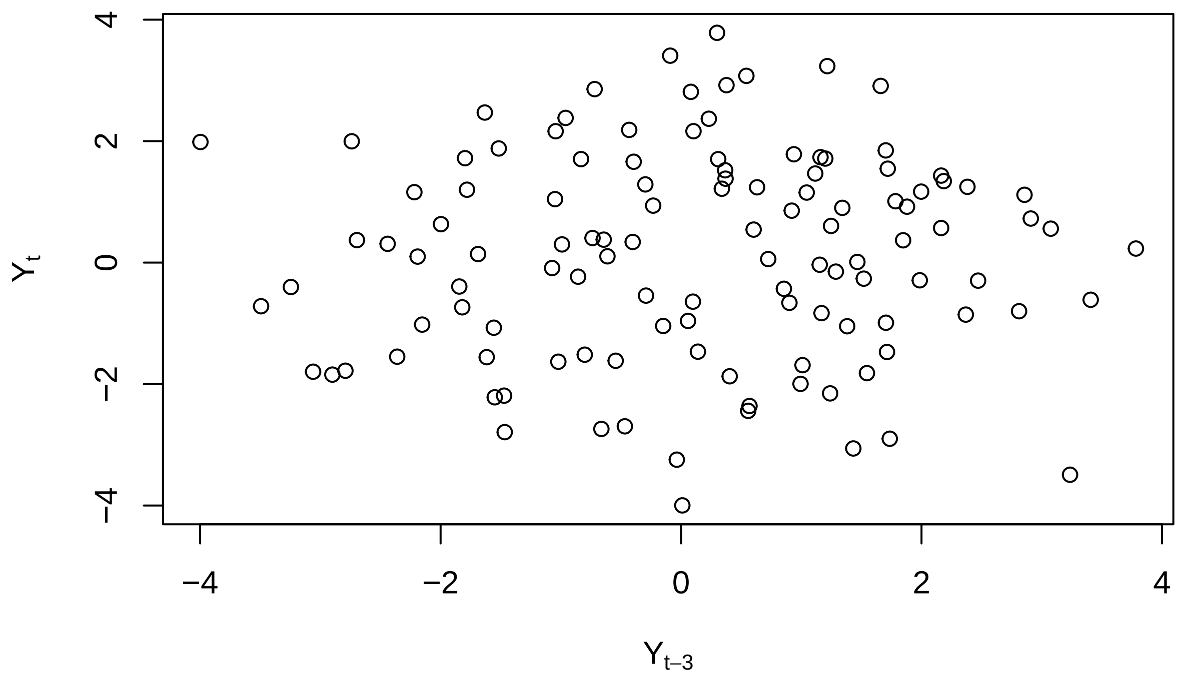
```r
#Exhibit 4.9
plot(y=ma2.s,x=zlag(ma2.s),ylab=expression(Y[t]),xlab=expression(Y[t-1]),type='p')
```



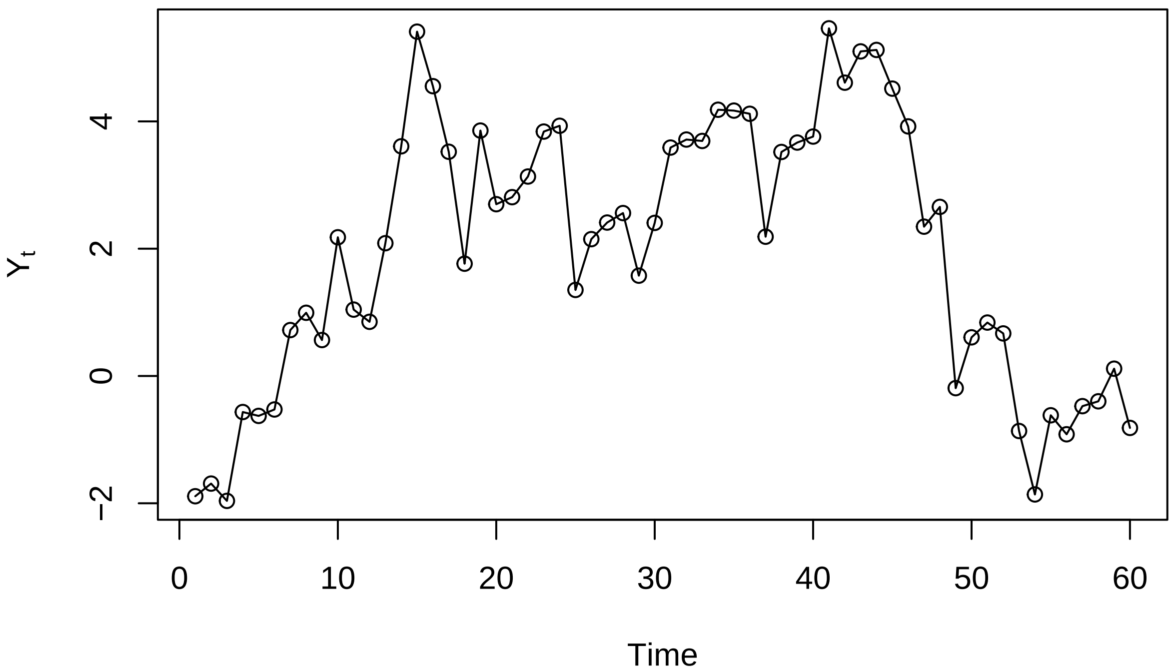

```
# Exhibit 4.10
plot(y=ma2.s,x=zlag(ma2.s,2),ylab=expression(Y[t]),xlab=expression(Y[t-2]),type='p')
```




```
# Exhibit 4.11
plot(y=ma2.s,x=zlag(ma2.s,3),ylab=expression(Y[t]),xlab=expression(Y[t-3]),type='p')
```

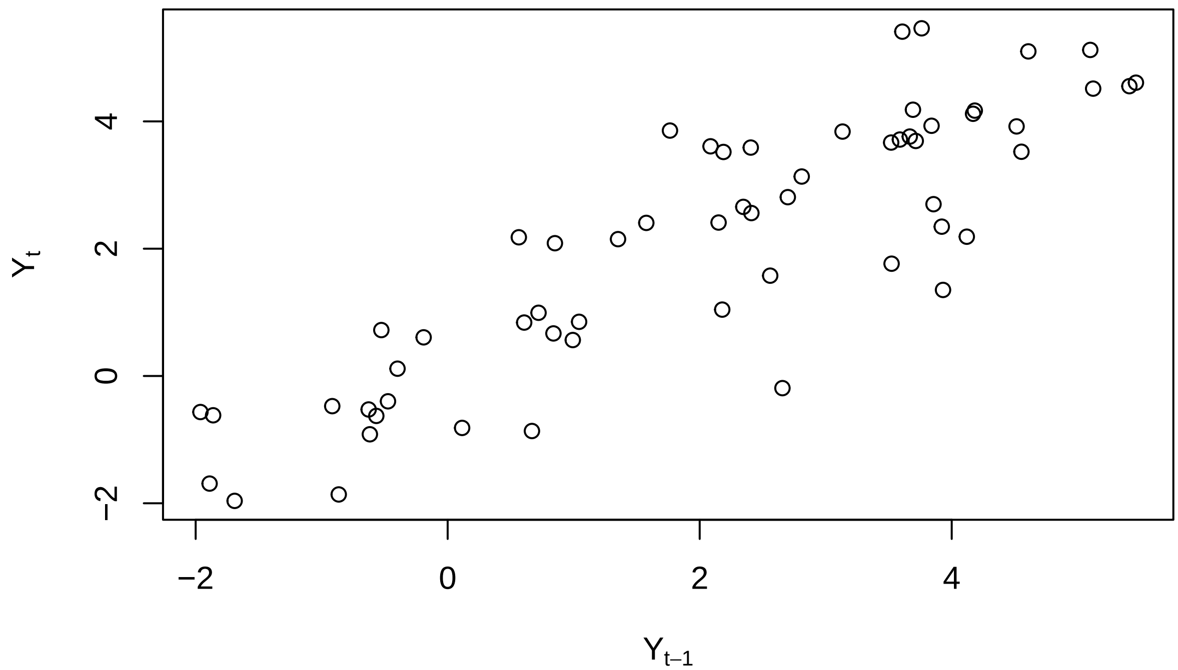



```r
# Exhibit 4.13
data(ar1.s)
plot(ar1.s,ylab=expression(Y[t]),type='o')
```

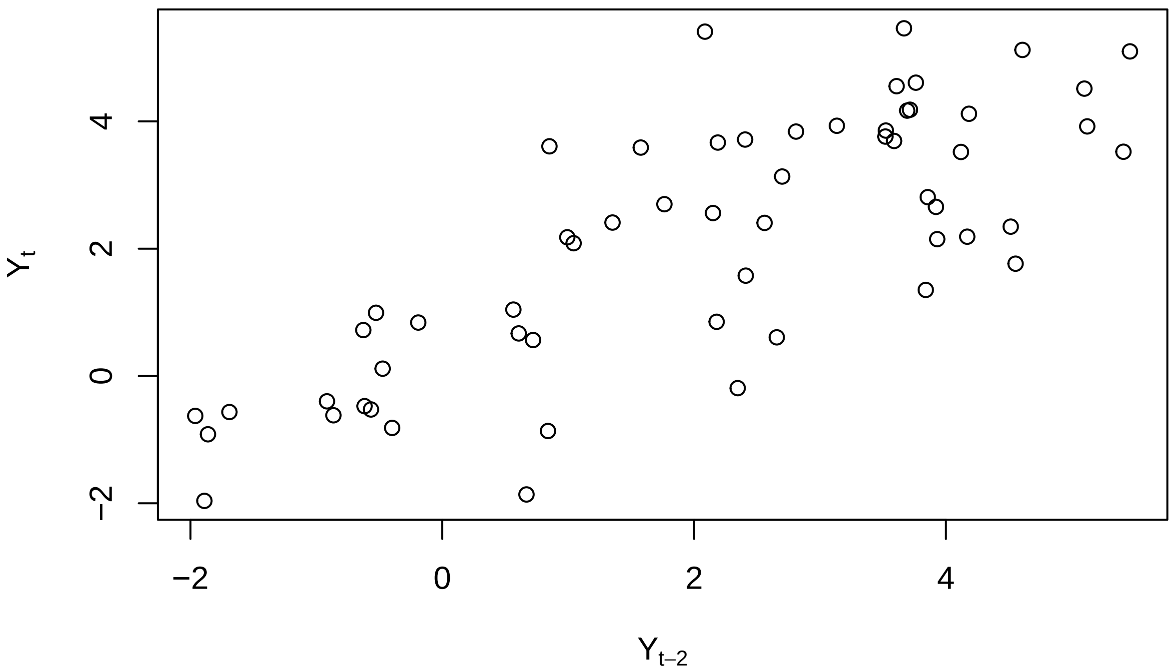



```r
y=arima.sim(model=list(ar=c(0.9)),n=100)
```

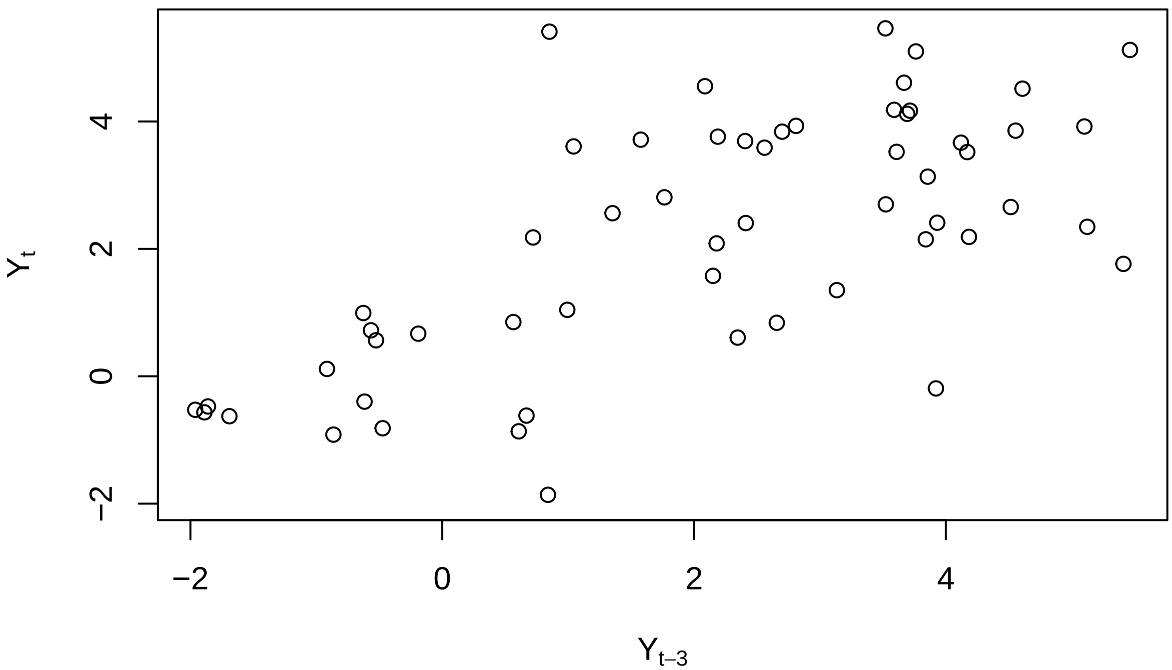
```r
# Exhibit 4.14
plot(y=ar1.s,x=zlag(ar1.s),ylab=expression(Y[t]),xlab=expression(Y[t-1]),type='p')
```



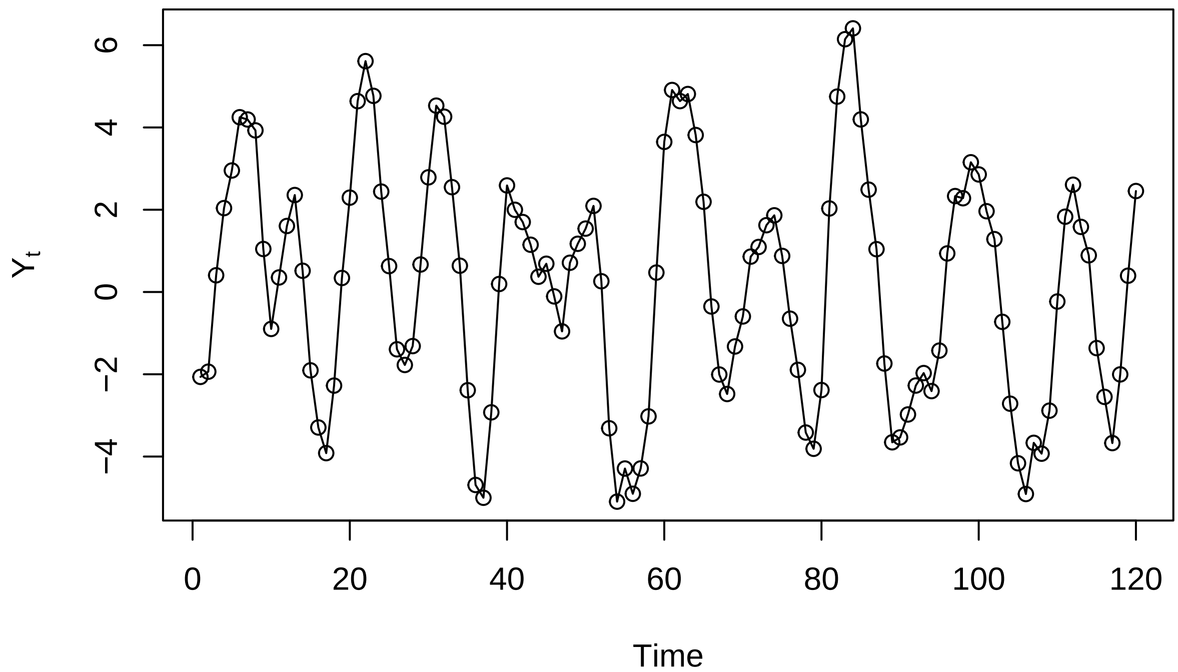

```r
# Exhibit 4.15
plot(y=ar1.s,x=zlag(ar1.s,2),ylab=expression(Y[t]),xlab=expression(Y[t-2]),type='p')
```




```
# Exhibit 4.16
plot(y=ar1.s,x=zlag(ar1.s,3),ylab=expression(Y[t]),xlab=expression(Y[t-3]),type='p')
```



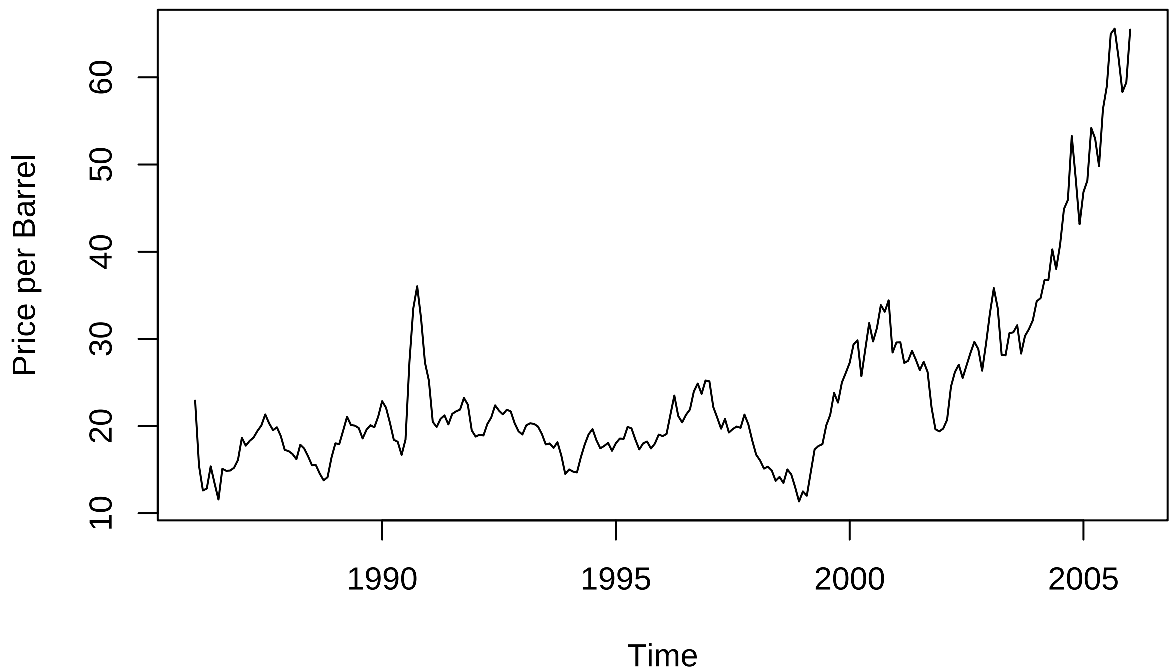

```r
# Exhibit 4.19
data(ar2.s)
plot(ar2.s,ylab=expression(Y[t]),type='o')
```



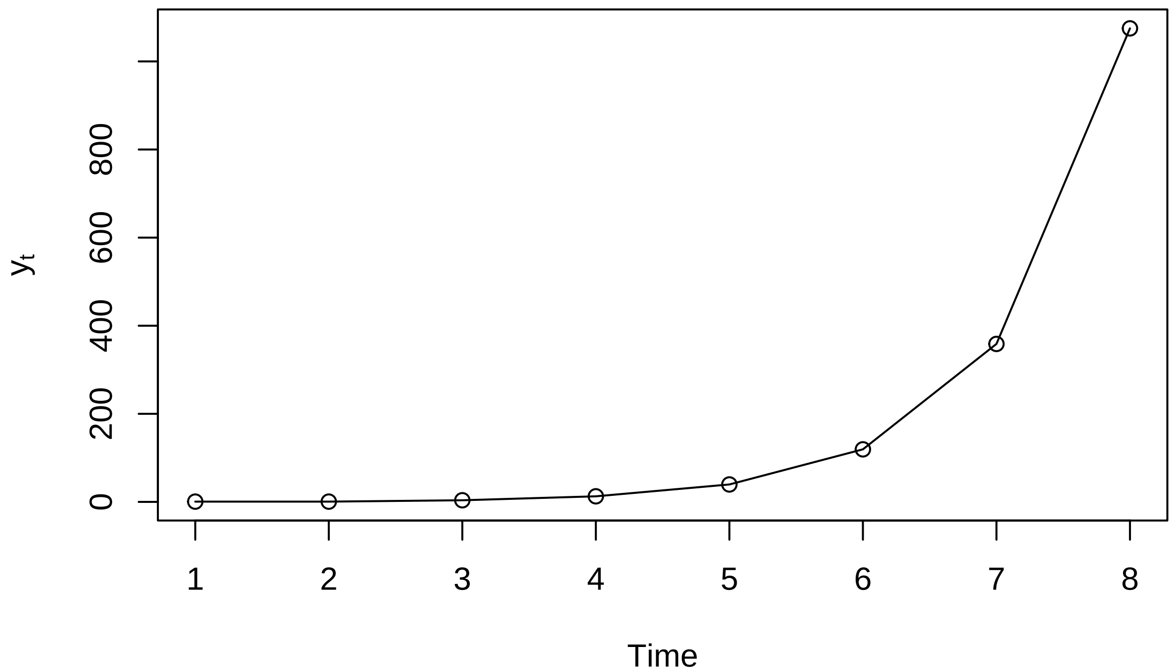

```
y=arima.sim(model=list(ar=c(1.5,-0.75)),n=100)
```

Chapter 5

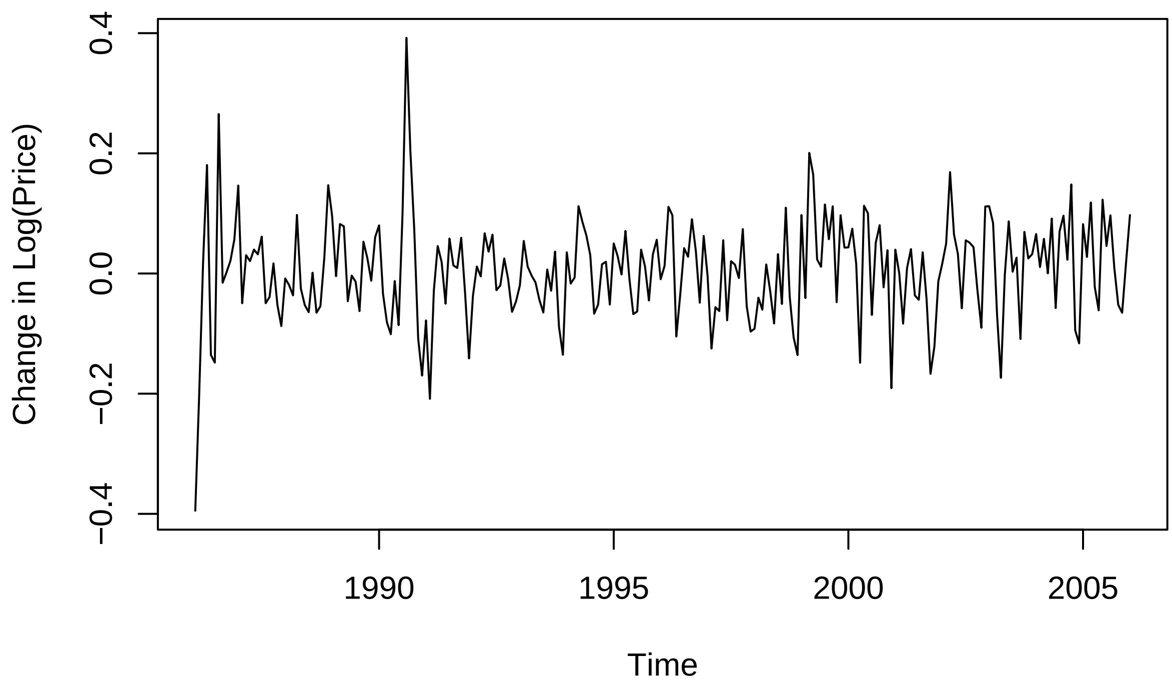
```
# Exhibit 5.1
data(oil.price)
plot(oil.price, ylab='Price per Barrel',type='l')
```



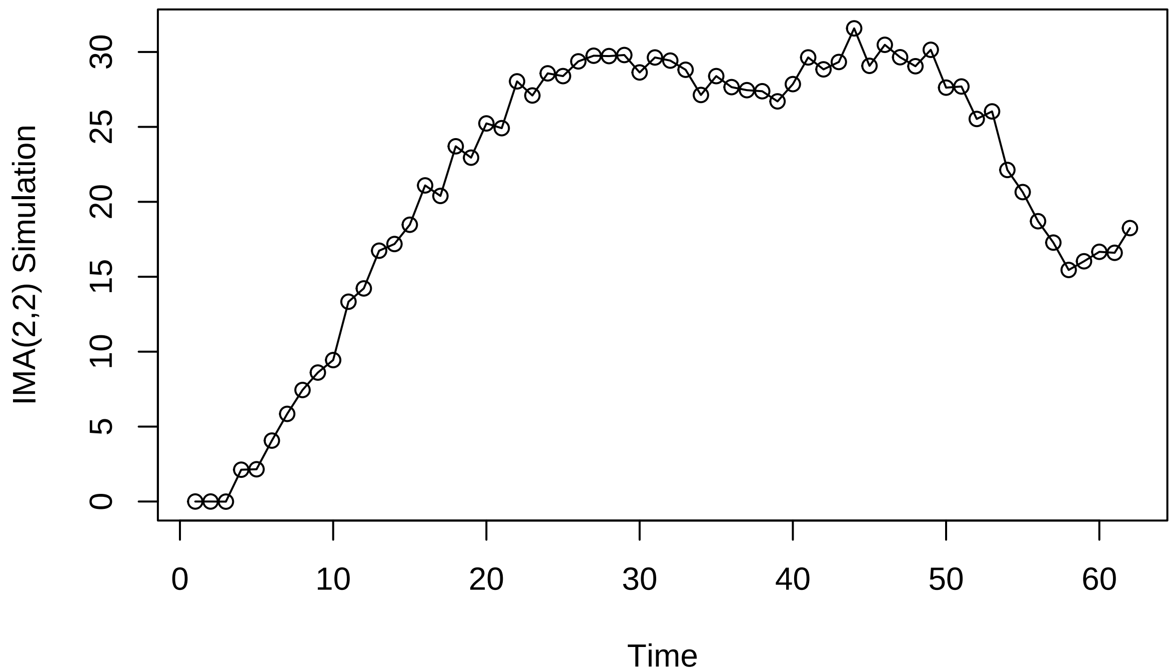

```
# Exhibit 5.3
data(explode.s)
plot(explode.s,ylab=expression(y[t]),type='o')
```



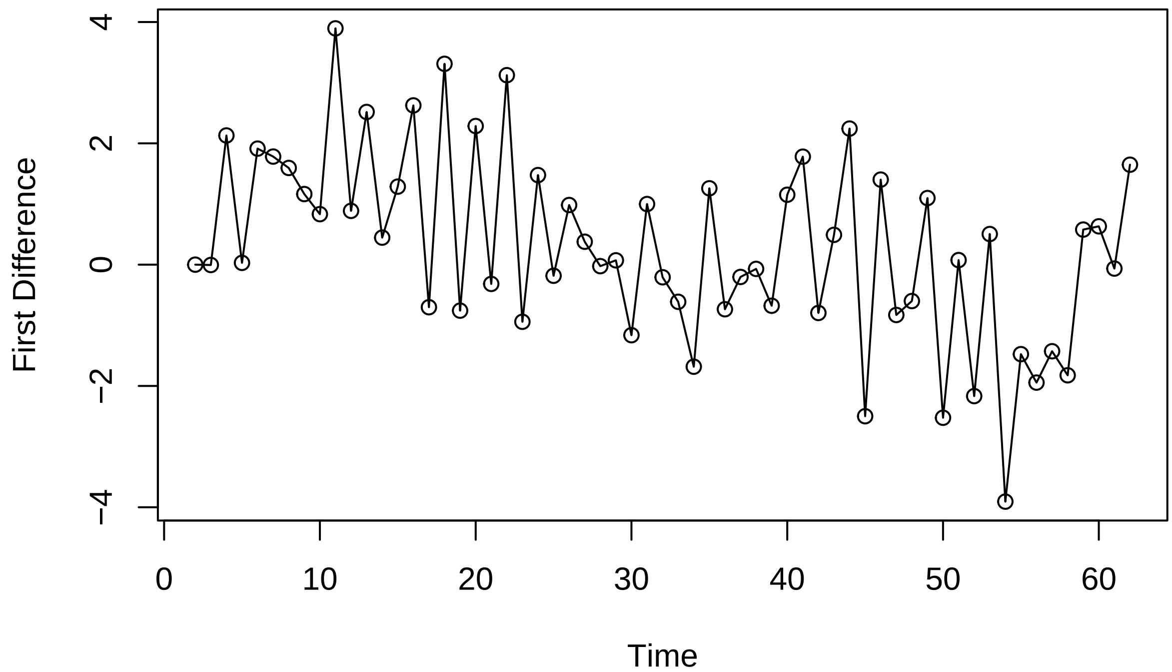

```r
# Exhibit 5.4
plot(diff(log(oil.price)),ylab='Change in Log(Price)',type='l')
```



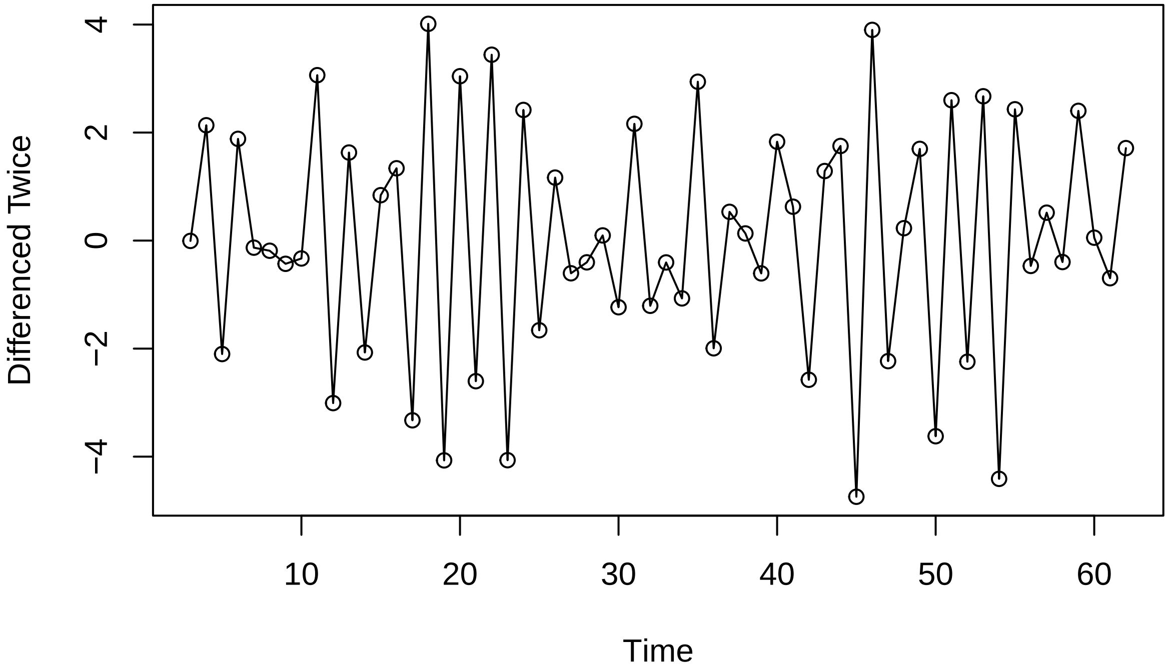

```
# Exhibit 5.5
data(ima22.s)
plot(ima22.s,ylab="IMA(2,2) Simulation",type='o')
```



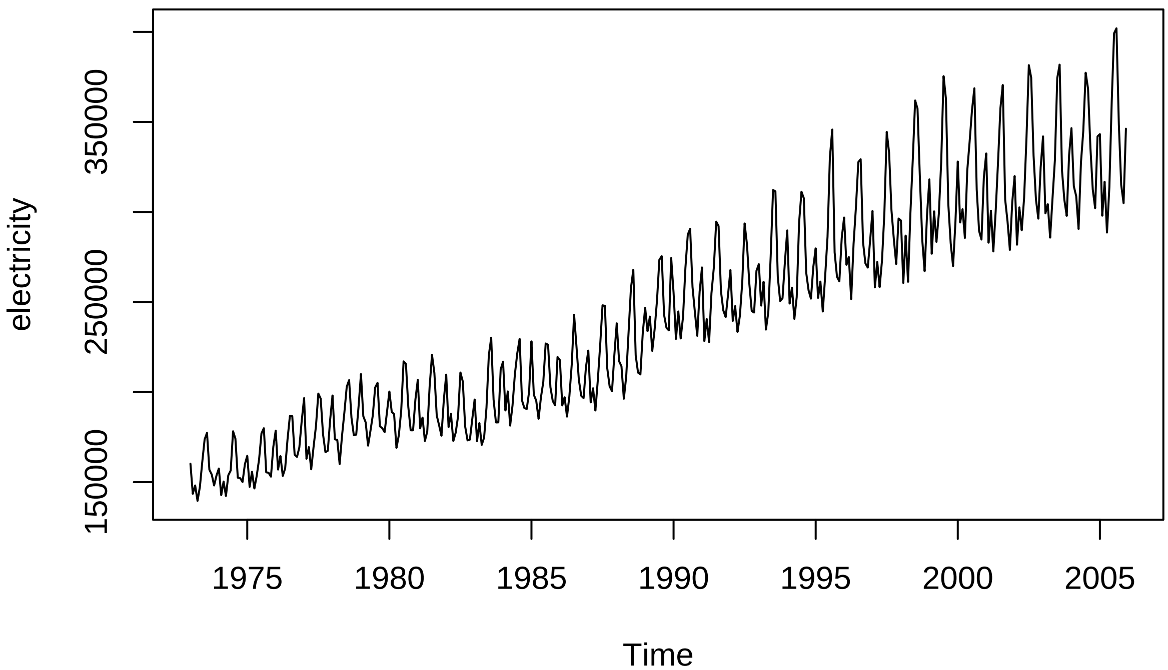

```
# Exhibit 5.6
plot(diff(ima22.s),ylab='First Difference',type='o')
```



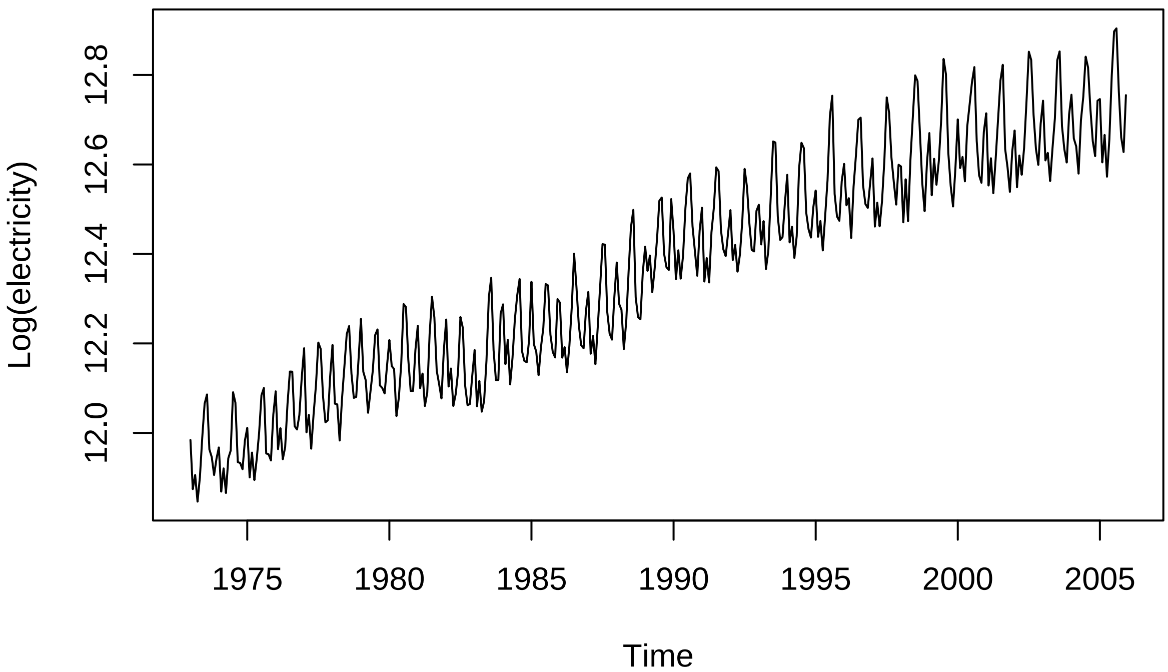

```r
# Exhibit 5.7
plot(diff(ima22.s,difference=2),ylab='Differenced Twice',type='o')
```



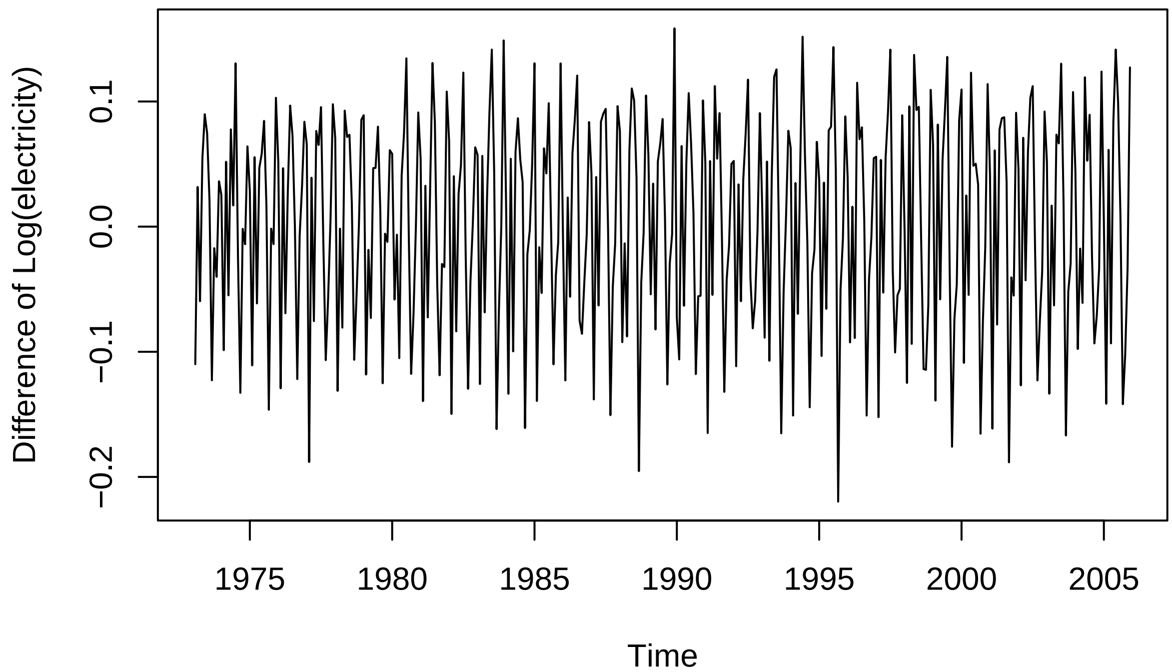

```r
# Exhibit 5.8
data(electricity)
plot(electricity)
```




```r
# Exhibit 5.9
plot(log(electricity),ylab='Log(electricity)')
```




```r
# Exhibit 5.10
plot(diff(log(electricity)),ylab='Difference of Log(electricity)')
```




```r
# Exhibit 5.11
BoxCox.ar(electricity)
```

```
## Warning in arima0(x, order = c(i, 0L, 0L), include.mean = demean): possible
## convergence problem: optim gave code = 1

## Warning in arima0(x, order = c(i, 0L, 0L), include.mean = demean): possible
## convergence problem: optim gave code = 1

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## convergence problem: optim gave code = 1

## Warning in arima0(x, order = c(i, 0L, 0L), include.mean = demean): possible
## convergence problem: optim gave code = 1
```
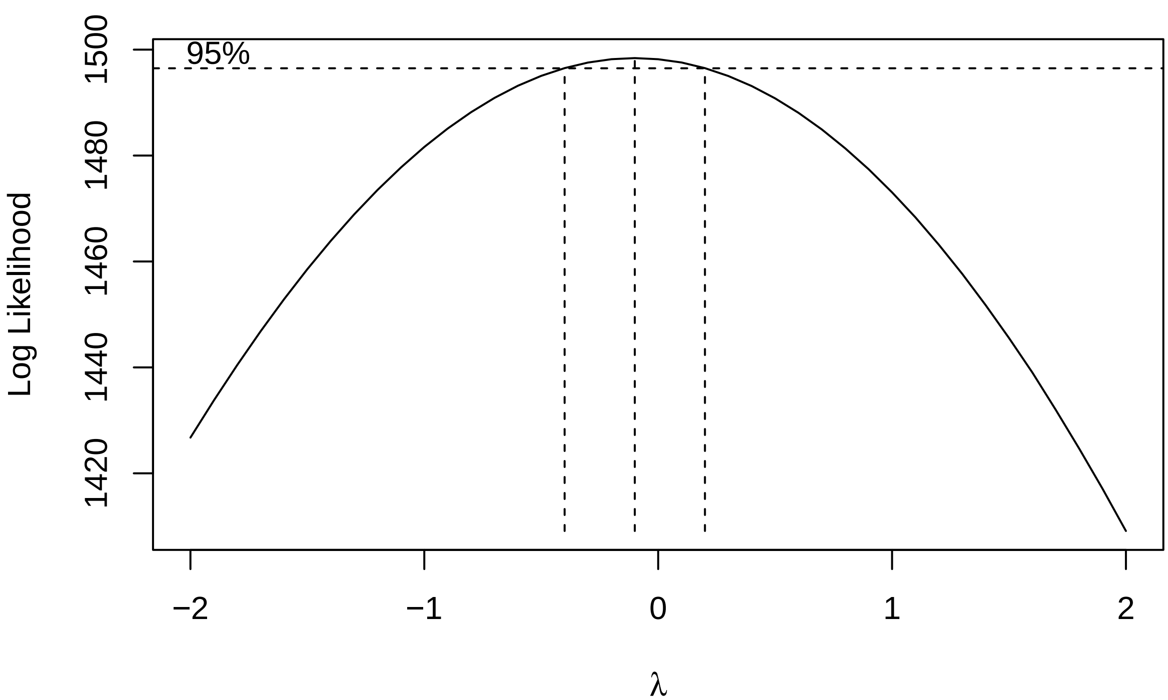
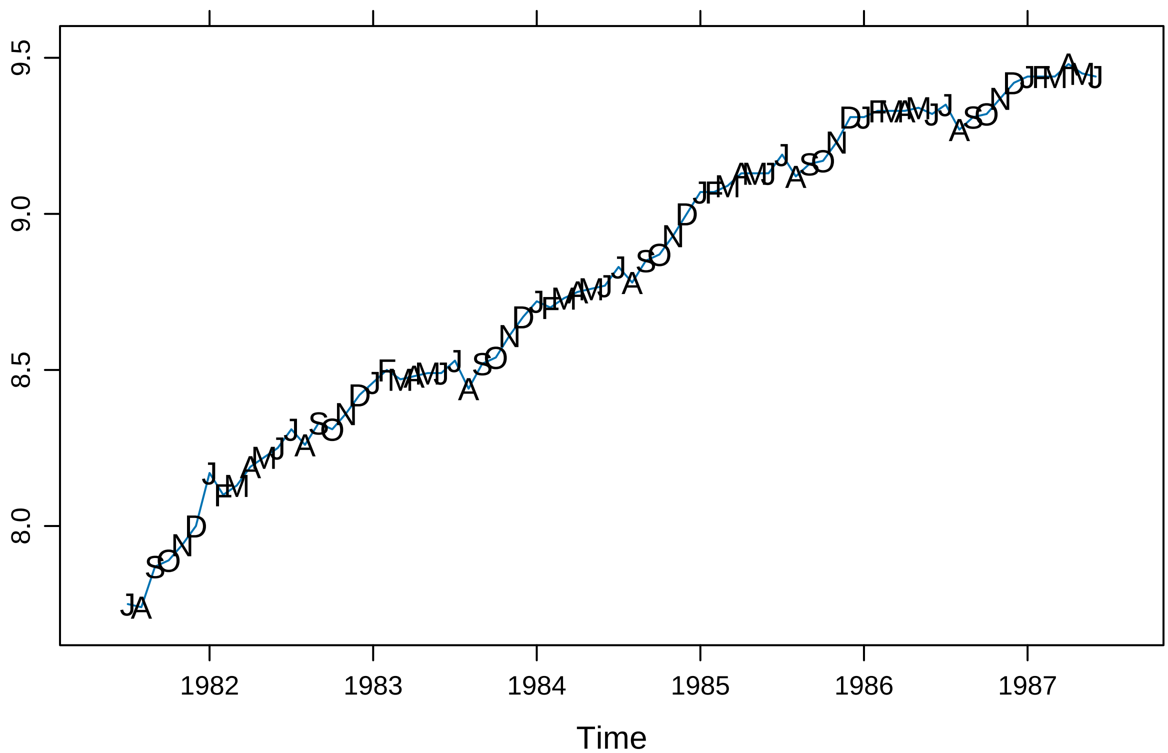

```
## Warning in arima0(x, order = c(i, 0L, 0L), include.mean = demean): possible
## convergence problem: optim gave code = 1

## Warning in arima0(x, order = c(i, 0L, 0L), include.mean = demean): possible
## convergence problem: optim gave code = 1

## Warning in arima0(x, order = c(i, 0L, 0L), include.mean = demean): possible
## convergence problem: optim gave code = 1

## Warning in arima0(x, order = c(i, 0L, 0L), include.mean = demean): possible
## convergence problem: optim gave code = 1

## Warning in arima0(x, order = c(i, 0L, 0L), include.mean = demean): possible
## convergence problem: optim gave code = 1

## Warning in arima0(x, order = c(i, 0L, 0L), include.mean = demean): possible
## convergence problem: optim gave code = 1

## Warning in arima0(x, order = c(i, 0L, 0L), include.mean = demean): possible
## convergence problem: optim gave code = 1
```



3.5

a

```
library(lattice)
months <- c("J", "A", "S", "O", "N", "D", "J", "F", "M", "A", "M", "J")
data("wages")
xyplot(wages, panel = function(x, y, ...)
{
panel.xyplot(x, y, ...)
panel.text(x, y, labels = months)
})
```



There is a positive trend with seasonality: August is a low-point for wages. Generally, there seems to be larger increases in the fall.

b

```
wages_fit1 <- lm(wages ~ time(wages))
summary(wages_fit1)
```

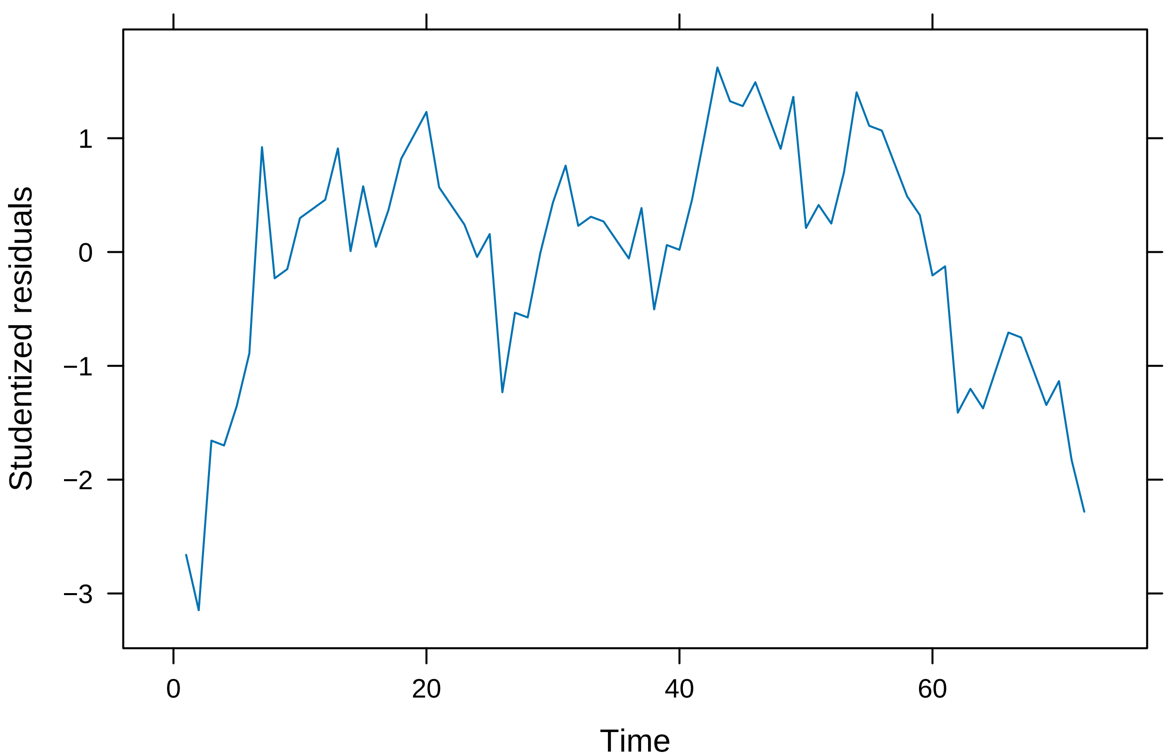
```
##
## Call:
```

```
## lm(formula = wages ~ time(wages))
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.23828 -0.04981  0.01942  0.05845  0.13136
##
## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept) -5.490e+02  1.115e+01  -49.24   <2e-16 ***
## time(wages)  2.811e-01  5.618e-03   50.03   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.08257 on 70 degrees of freedom
## Multiple R-squared:  0.9728, Adjusted R-squared:  0.9724
## F-statistic:  2503 on 1 and 70 DF,  p-value: < 2.2e-16
```

```r
wages_rst <- rstudent(wages_fit1)
```

c

```r
xyplot(wages_rst ~ time(wages_rst), type = "l",
       xlab = "Time", ylab = "Studentized residuals")
```



We still seem to have autocorrelation related to the time and not white noise.

d

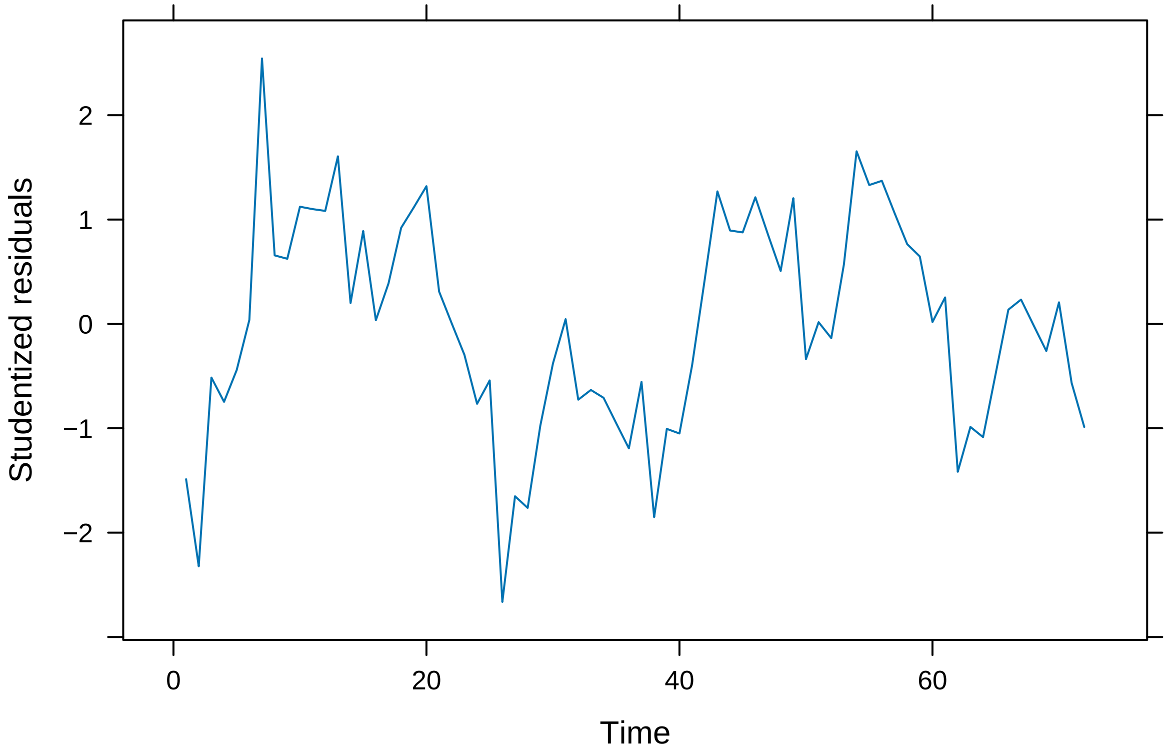
```r
wages_fit2 <- lm(wages ~ time(wages) + I(time(wages)^2))
summary(wages_fit2)
```

```
##
## Call:
## lm(formula = wages ~ time(wages) + I(time(wages)^2))
##
## Residuals:
##       Min        1Q    Median        3Q       Max
## -0.148318 -0.041440  0.001563  0.050089  0.139839
##
## Coefficients:
##                   Estimate Std. Error t value Pr(>|t|)
## (Intercept)      -8.495e+04  1.019e+04  -8.336 4.87e-12 ***
## time(wages)       8.534e+01  1.027e+01   8.309 5.44e-12 ***
## I(time(wages)^2) -2.143e-02  2.588e-03  -8.282 6.10e-12 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.05889 on 69 degrees of freedom
## Multiple R-squared:  0.9864, Adjusted R-squared:  0.986
## F-statistic:  2494 on 2 and 69 DF,  p-value: < 2.2e-16
```

```r
wages_rst2 <- rstudent(wages_fit2)
```

e

```r
xyplot(wages_rst2 ~ time(wages_rst), type = "l",
       xlab = "Time", ylab = "Studentized residuals")
```

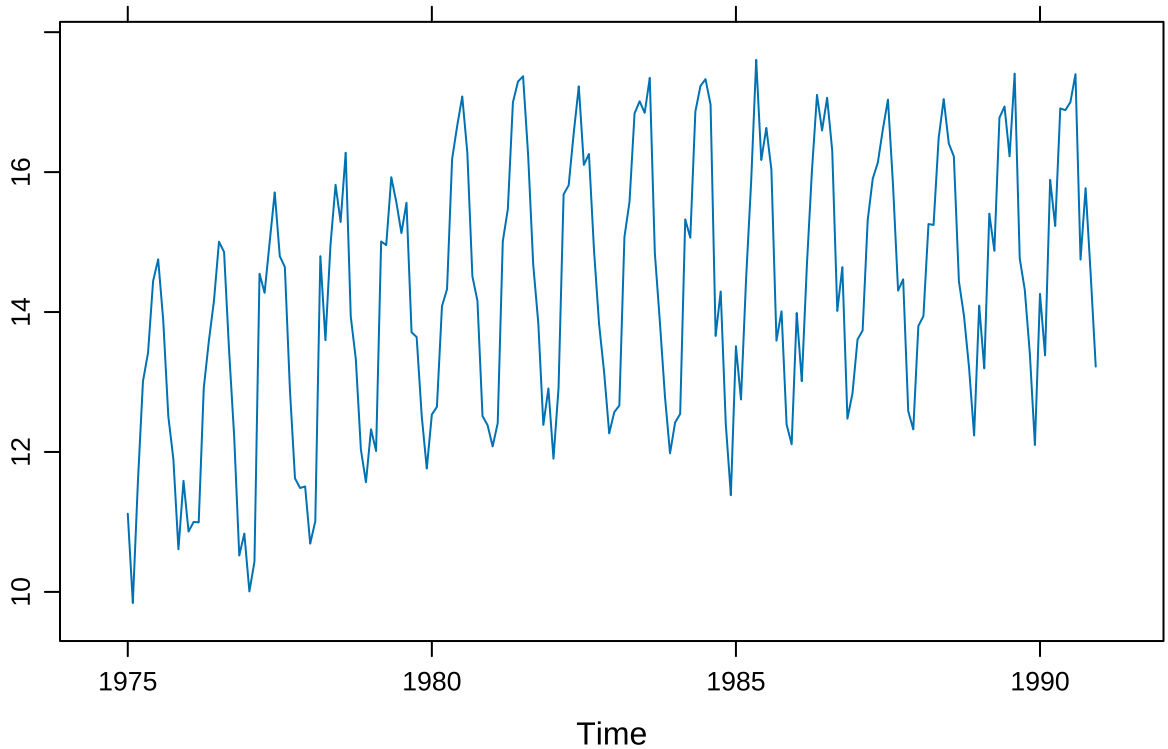


This looks more like random noise but there is still clear autocorrelation between the fitted residuals that we have yet to capture in our model.

3.6

a

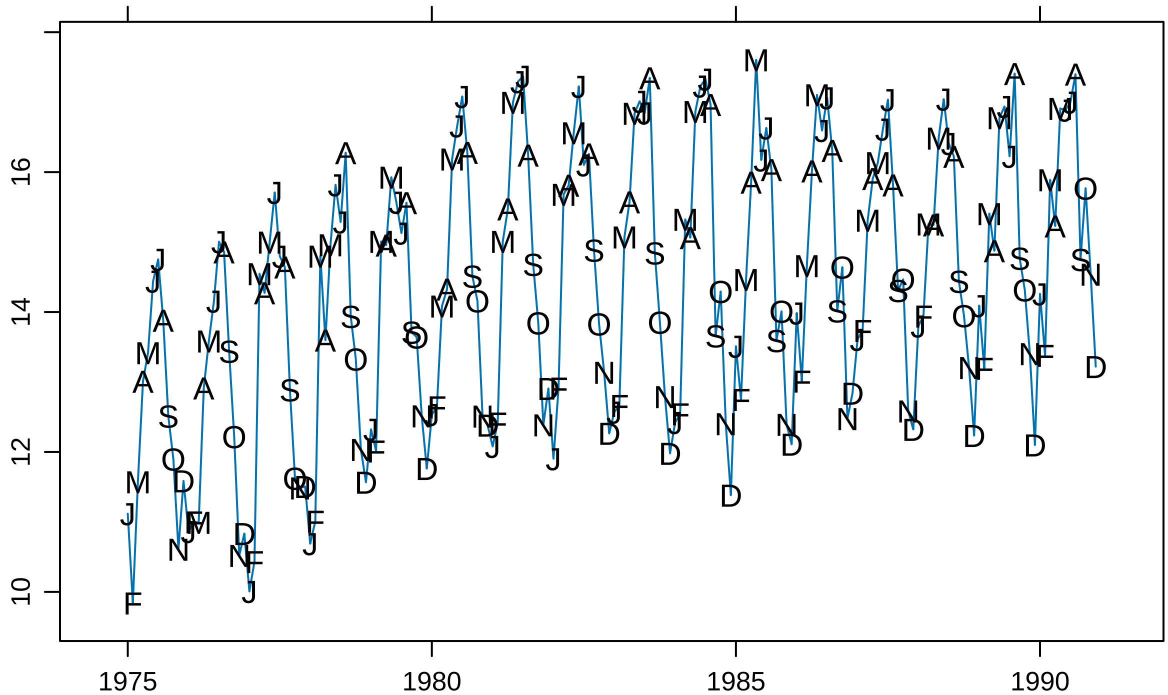
```
data(beersales)
xyplot(beersales)
```



Clear seasonal trends. There is an initial positive trend from 1975 to around 1981 that then levels out.

b

```r
months <- c("J", "F", "M", "A", "M", "J", "J", "A", "S", "O", "N", "D")

xyplot(beersales,
       panel = function(x, y, ...) {
         panel.xyplot(x, y, ...)
         panel.text(x, y, labels = months)
       })
```



It is now evident that the peaks are in the warm months and the slump in the winter and fall months. December is a particular low point, while May, June, and July seem to be the high points.

c

```r
library(pander)
beer_fit1 <- lm(beersales ~ season(beersales))
pander(summary(beer_fit1))
```

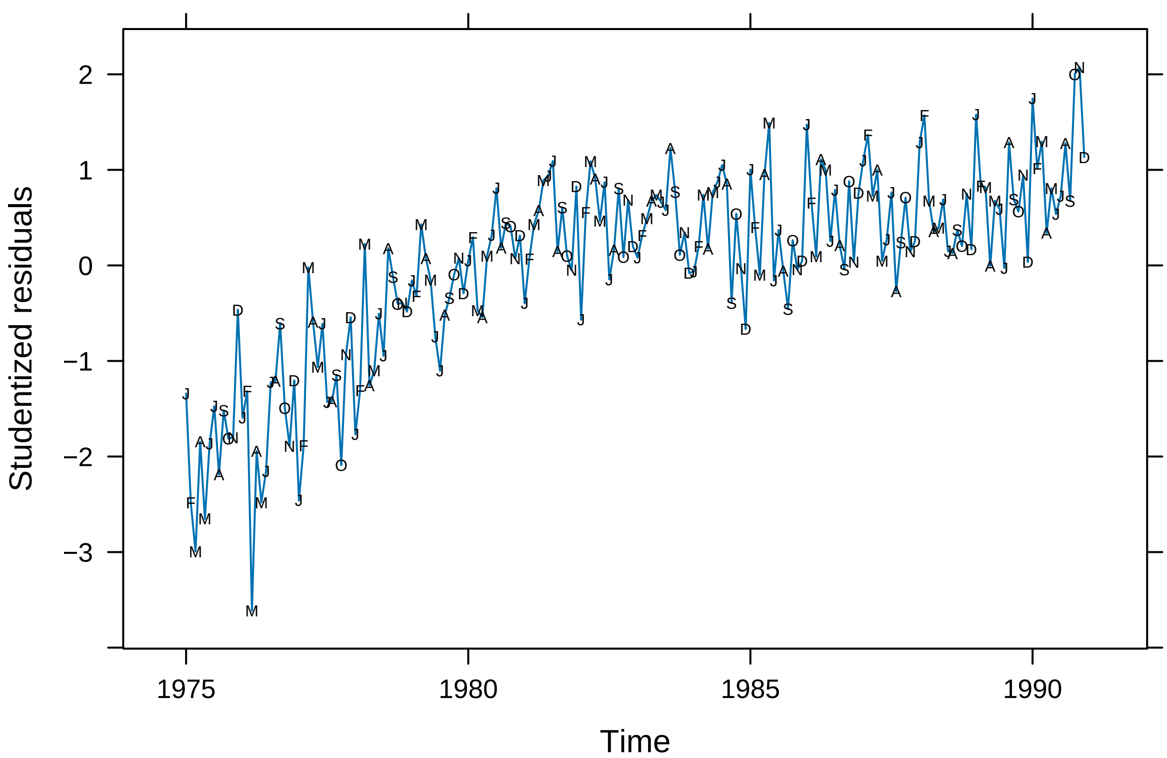
	Estimate	Std. Error	t value	Pr(>|t|)
(Intercept)	12.49	0.2639	47.31	1.786e-103
season(beersales)February	-0.1426	0.3732	-0.382	0.7029
season(beersales)March	2.082	0.3732	5.579	8.771e-08
season(beersales)April	2.398	0.3732	6.424	1.151e-09
season(beersales)May	3.599	0.3732	9.643	5.322e-18
season(beersales)June	3.85	0.3732	10.31	6.813e-20
season(beersales)July	3.769	0.3732	10.1	2.812e-19
season(beersales)August	3.609	0.3732	9.669	4.494e-18
season(beersales)September	1.573	0.3732	4.214	3.964e-05
season(beersales)October	1.254	0.3732	3.361	0.0009484
season(beersales)November	-0.04797	0.3732	-0.1285	0.8979
season(beersales)December	-0.4231	0.3732	-1.134	0.2585

Table 2: Fitting linear model: beersales ~ season(beersales) All comparisons are made against january. The model helpfully explains approximately 0.71 of the variance and is statistically significant. Most of the factors are significant (mostly the winter months as expected).

Observations	Residual Std. Error	R^2	Adjusted R^2
192	1.056	0.7103	0.6926

d

```
xyplot(rstudent(beer_fit1) ~ time(beersales), type = "l",
       xlab = "Time", ylab = "Studentized residuals",
       panel = function(x, y, ...) {
         panel.xyplot(x, y, ...)
         panel.xyplot(x, y, pch = as.vector(season(beersales)), col = 1)
       })
```



Looking at the residuals in the figure, We don't have a good fit to our data; in particular, wee're not capturing the long-term trend.

e

```
beer_fit2 <- lm(beersales ~ season(beersales) + time(beersales) +
                  I(time(beersales) ^ 2))
pander(summary(beer_fit2))
```

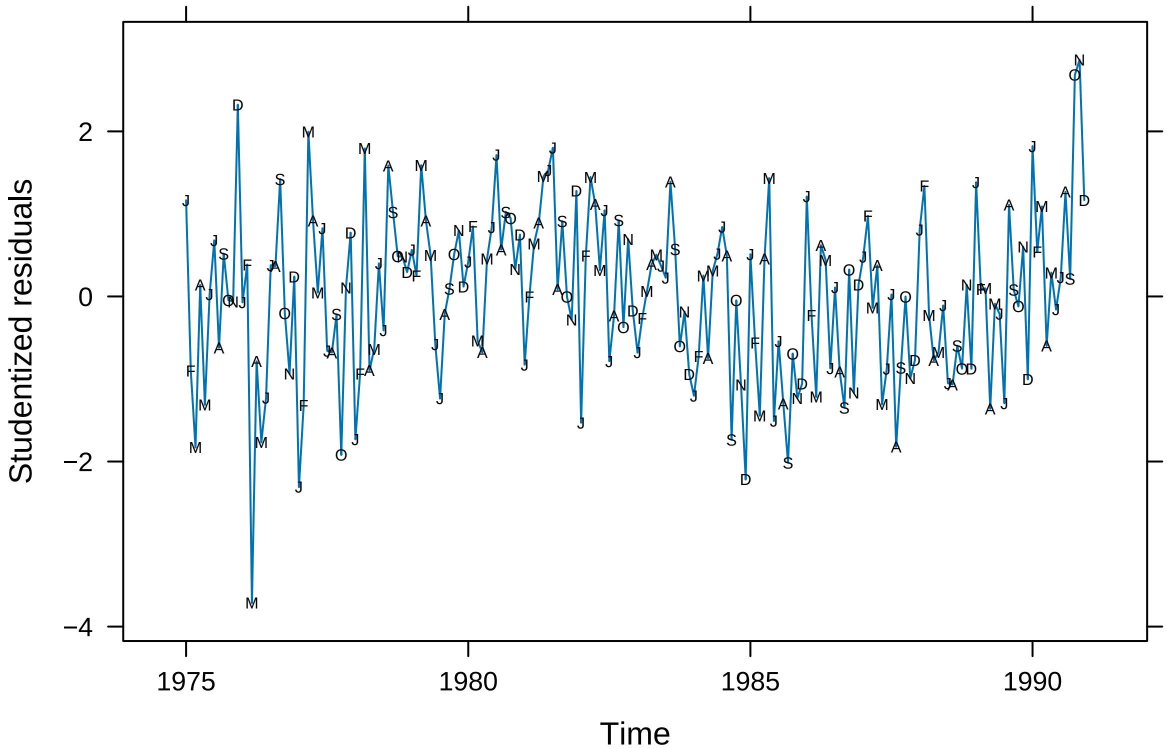
	Estimate	Std. Error	t value	Pr(>|t|)
(Intercept)	-71498	8791	-8.133	6.932e-14
season(beersales)February	-0.1579	0.209	-0.7554	0.451
season(beersales)March	2.052	0.209	9.818	1.864e-18
season(beersales)April	2.353	0.209	11.26	1.533e-22
season(beersales)May	3.539	0.209	16.93	6.063e-39
season(beersales)June	3.776	0.209	18.06	4.117e-42
season(beersales)July	3.681	0.209	17.61	7.706e-41
season(beersales)August	3.507	0.2091	16.78	1.698e-38
season(beersales)September	1.458	0.2091	6.972	5.89e-11
season(beersales)October	1.126	0.2091	5.385	2.268e-07
season(beersales)November	-0.1894	0.2091	-0.9059	0.3662
season(beersales)December	-0.5773	0.2092	-2.76	0.00638
time(beersales)	71.96	8.867	8.115	7.703e-14
I(time(beersales)^2)	-0.0181	0.002236	-8.096	8.633e-14

Table 4: Fitting linear model: beersales ~ season(beersales) + time(beersales) + I(time(beersales)^2) This model fits the data better, explaining roughly 0.91 of the variance.

Observations	Residual Std. Error	R^2	Adjusted R^2
192	0.5911	0.9102	0.9036

f

```
xyplot(rstudent(beer_fit2) ~ time(beersales), type = "l",
       xlab = "Time", yla = "Studentized residuals",
       panel = function(x, y, ...) {
         panel.xyplot(x, y, ...)
         panel.xyplot(x, y, pch = as.vector(season(beersales)), col = 1)
       })
```



Many of the values are still not being predicted successfully but at least we're able to model the long term trend better.

3.12

a

```
beer_quad_seasonal <- lm(beersales ~ time(beersales) + I(time(beersales)^2) +
                           season(beersales))
beer_resid <- rstudent(beer_quad_seasonal)
```

b

```
runs(beer_resid)
```

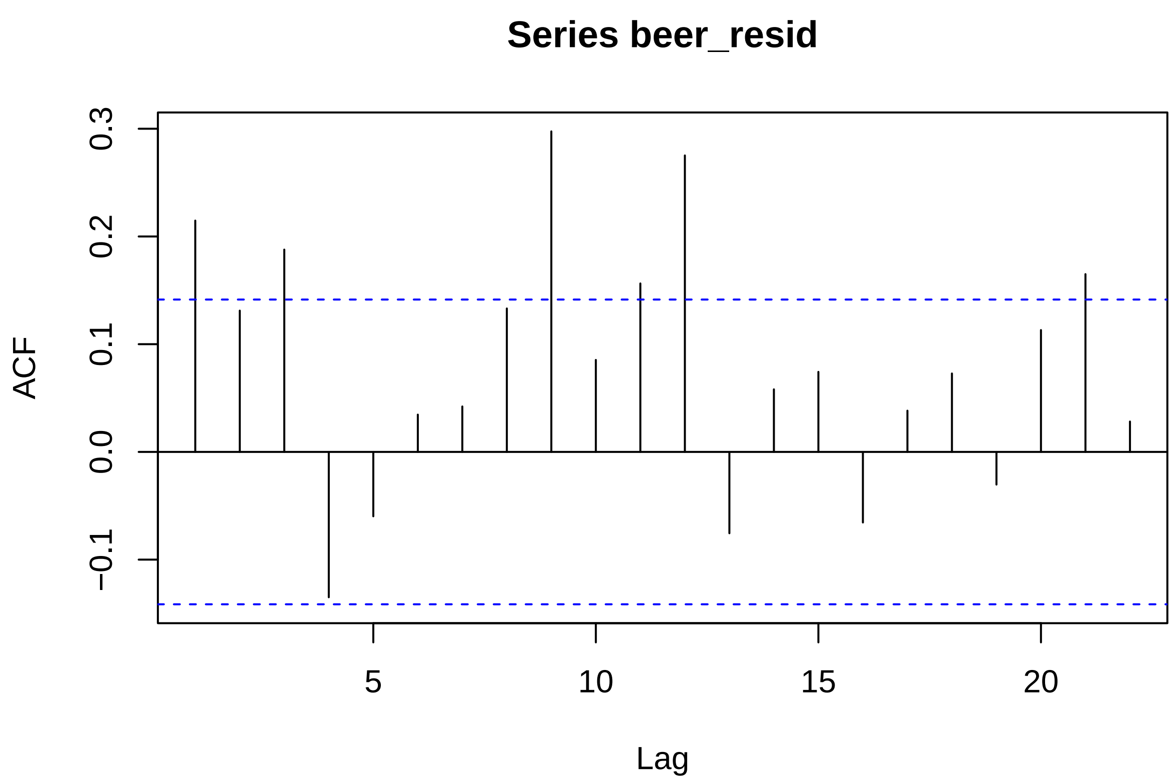
```
## $pvalue
## [1] 0.0127
##
## $observed.runs
## [1] 79
##
```

```
## $expected.runs
## [1] 96.625
##
## $n1
## [1] 90
##
## $n2
## [1] 102
##
## $k
## [1] 0
```

The test is significant (p = 0.01).

c

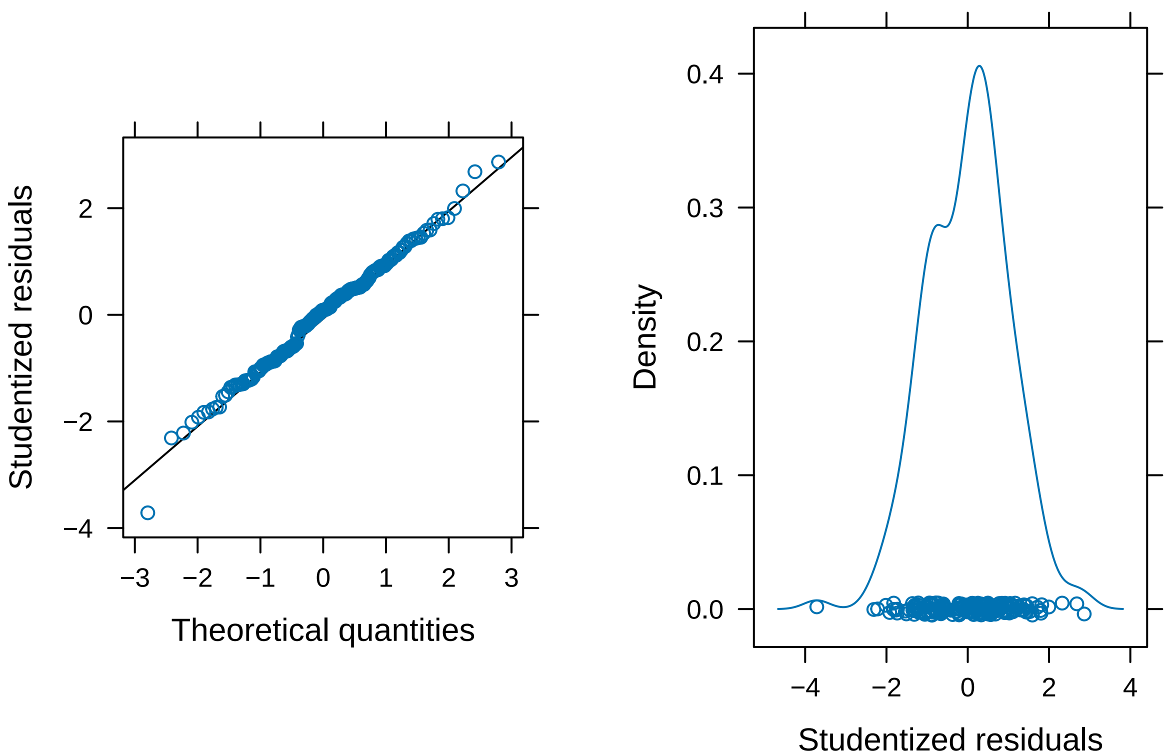
```r
acf(beer_resid)
```



Correlations are significant for several of the lags, leading us to question independence.

d

```
figa <-
  qqmath(beer_resid, xlab = "Theoretical quantities",
         asp = 1,
         ylab = "Studentized residuals",
         panel = function(x, ...) {
           panel.qqmathline(x, ...)
           panel.qqmath(x, ...)
         })

figb <- densityplot(beer_resid, xlab = "Studentized residuals")
gridExtra::grid.arrange(figa, figb, ncol = 2)
```



Question-3:-

4.1) Sol) Given,

$$Y_t = 5 + e_t - \frac{1}{2} e_{t-1} + \frac{1}{4} e_{t-2}$$

We will assume that $\{e_t\}$ is a white noise process with mean zero and variance σ^2

$$\gamma_k = Cov(Y_t, Y_{t-k})$$

a) γ_0 (Variance):- The auto covariance at lag 0 is the variance of Y_t

$$\gamma_0 = Var(Y_t) = Var\left(5 + e_t - \frac{1}{2} e_{t-1} + \frac{1}{4} e_{t-2}\right)$$

$$= Var\left[1 + \left(\frac{1}{2}\right)^2 + \left(\frac{1}{4}\right)^2\right]$$

$$= \sigma^2\left[1 + \frac{1}{4} + \frac{1}{16}\right] = \frac{21}{16}\sigma^2$$

$$\boxed{\gamma_0 = \frac{21}{16}\sigma^2}$$

b) Autocovariance at Lag 1:-

$$Cov(Y_t, Y_{t-1}) = Cov\left(5 + e_t - \frac{1}{2} e_{t-1} + \frac{1}{4} e_{t-2}, 5 + e_{t-1} - \frac{1}{2} e_{t-2} + \frac{1}{4} e_{t-3}\right)$$

Since e_t terms are white noise the only terms with non-zero covariance are those where indices match.

$$Cov(Y_t, Y_{t-1}) = Cov\left(-\frac{1}{2} e_{t-1}, e_{t-1}\right) + Cov\left(\frac{1}{4} e_{t-2}, -\frac{1}{2} e_{t-2}\right)$$

$$= -\frac{1}{2} Var(e_{t-1}) - \frac{1}{8} Var(e_{t-2})$$

$$\boxed{Cov(Y_t, + Y_{t-1}) = -\frac{5}{8}\sigma_e^2} = \gamma_1$$

c) Auto covariance at Lag 2:-

$$Cov(Y_t, Y_{t-2}) = Cov\left(5 + e_t - \frac{1}{2} e_{t-1} + \frac{1}{4} e_{t-2}, 5 + e_{t-2} - \frac{1}{2} e_{t-3} + \frac{1}{4} e_{t-4}\right)$$

$$= Cov\left(\frac{1}{4} e_{t-2}, e_{t-2}\right) = \frac{1}{4} Var(e_{t-2})$$

$$\boxed{Cov(Y_t, Y_{t-2}) = \frac{1}{4}\sigma_e^2} = \gamma_2$$

d) Autocovariance at Lag3 (or) higher:-

There will be no overlapping between e_t terms so $\boxed{\gamma_k = 0}$ for $k \geq 3$

Auto correlation function (ρ_k):-

$$\rho_k = \frac{\gamma_k}{\gamma_0}$$

$$\therefore \rho_0 = \frac{\gamma_0}{\gamma_0} = 1$$

$$\rho_1 = \frac{\gamma_1}{\gamma_0} = \frac{-\frac{5}{8}}{\frac{21}{16}} \Rightarrow \frac{-10}{21}$$

$$\rho_2 = \frac{\gamma_2}{\gamma_0} = \frac{\frac{1}{4}}{\frac{21}{16}} \Rightarrow \frac{4}{21}$$

$$\rho_k = \frac{\gamma_k}{\gamma_0} \text{ where } k \geq 3 \Rightarrow \rho_k = 0$$

$$\rho_k = \begin{cases} 1 & k=0 \\ \frac{-10}{21} & k=1 \\ \frac{4}{21} & k=2 \\ 0 & k \geq 3. \end{cases}$$

4.6) Sol) Given $w_t = \nabla Y_t = Y_t - Y_{t-1}$ where Y_t is stationary process with ϕ & σ_e^2

a) Covariance of ∇Y_t:-

$$Cov(\nabla Y_t, \nabla Y_{t-k}) = Cov(Y_t - Y_{t-1}, Y_{t-k} - Y_{t-k-1})$$

$$= Cov(Y_t, Y_{t-k}) - Cov(Y_{t-1}, Y_{t-k}) - Cov(Y_t, Y_{t-k-1}) + Cov(Y_{t-1}, Y_{t-k-1})$$

$$= \frac{\sigma_e^2}{1-\phi^2}(\phi^k - \phi^{k-1} - \phi^{k+1} + \phi^k) \quad \because Var(w_t)$$

$$= \frac{\sigma_e^2}{1-\phi^2}\phi^{k-1}(2\phi - \phi - 1)$$

$$\Rightarrow \boxed{Cov(\nabla Y_t, \nabla Y_{t-k}) = -\sigma_e^2 \cdot \frac{(1-\phi)}{1+\phi}\phi^{k-1}}$$

b) Variance of w_t:-

$$\gamma_w(0) = Var(w_t) = Var(Y_t - Y_{t-1})$$

$$= Var(\phi Y_{t-1} + e_t - Y_{t-1})$$

$$= (\phi-1)^2 \cdot Var(Y_{t-1}) + Var(e_t)$$

$$= \frac{\sigma_e^2}{1-\phi^2}(\phi^2 - 2\phi + 1) + \sigma_e^2$$

$$\Rightarrow \boxed{Var(w_t) = \frac{2\sigma_e^2}{1+\phi}}$$

4.7) sol)

a) MA(1) – Moving Average of order 1

$$Y_t = \mu + e_t + \theta e_{t-1}$$ where e_t is white noise

ACF Characteristics:-

* ACF has single non zero lag at $k=1$, given by $e_1 = \frac{\theta}{1+\theta^2}$
* for lags $k \geq 2$, the autocorrelation is zero
* This "cutoff" after lag 1 is a key indicator of an MA(1) process. So there will be only correlation at lag 1.

b) MA(2) – Moving Average of order 2.

$$Y_t = \mu + e_t + \theta_1 e_{t-1} + \theta_2 e_{t-2}$$

ACF Characteristics:-

* The ACF has non zero values at lags $k=1$ and $k=2$ and then drops to zero for $k \geq 3$.
* The values at lags 1 and 2 depends on parameters θ_1 and θ_2 and the cutoff after lag 2 is characteristic of an MA(2) process.
* So there will be only autocorrelation at lag 1 and lag 2, shape of process depends on values of coefficients.

c) AR(1) – Auto regressive of order 1

$$Y_t = \phi Y_{t-1} + e_t$$

ACF Characteristics:-

* The ACF decreases exponentially as k increases, either decaying or oscillating depends on ϕ.
  → If $\phi > 0$, the ACF decays gradually with positive values.
  → If $\phi < 0$, the ACF oscillates with alternating signs.
* The decay rate depends on the magnitude of ϕ; higher values (close to 1 or −1) produce slower decay.
* The absence of a sharp cutoff in the ACF is a key indicator of an AR process.
* So there will be exponentially decaying correlation from lag 0.

d) AR(2) – Auto regressive of Order 2

$$Y_t = \phi_1 Y_{t-1} + \phi_2 Y_{t-2} + e_t$$

ACF Characteristics:-

* The ACF may exhibit damped exponential decay or damped oscillations depending on the values of ϕ_1 and ϕ_2
* If the roots of the characteristic eqⁿ are complex, the ACF oscillates, if they are real the ACF decays more smoothly.
* Like AR(1), the ACF does not cut off but decays gradually, with the exact shape influenced by the parameters.

e) ARMA(1, 1):- Autoregressive Moving Average of Order(1,1)

$$Y_t = \phi Y_{t-1} + e_t + \theta e_{t-1})$$

ACF characteristics:-

* The ACF exhibits a mixed behaviour, it initially resembles the decay pattern of AR(1) process but then cuts off more quickly than a pure AR(1) due to the MA component.
* The exact shape of ACF depends on the values of ϕ and θ for moderate values, the ACF might show an exponential decay with some influence from MA term, resulting in a faster decline.
* The combined AR and MA characteristics make the ARMA(1,1) ACF more flexible, allowing for a range of decay patterns.
* So there will be exponentially decaying correlation from lag 1.

4.11 sol) Let $Y_t = 0.8Y_{t-1} + e_t + 0.7e_{t-1} + 0.6e_{t-2}$

a) covariance of Y_t at lag $k > 2$

$$Cov(Y_t, Y_{t-k}) = E[(0.8Y_{t-1} + e_t + 0.7e_{t-1} + 0.6e_{t-2}) Y_{t-k}] - E[Y_t]\,E[Y_{t-k}]$$

$$= 0.8E[Y_{t-1}Y_{t-k}] + E[Y_{t-k}e_t] + 0.7E(e_{t-1}Y_{t-k}) + 0.6E(e_{t-2}Y_{t-k})$$

$$= 0.8\,Cov(Y_t, Y_{t-k+1})$$

$$= 0.8\gamma_{k-1}$$

$$\boxed{\therefore Cov(Y_t, Y_{t-k}) = 0.8\gamma_{k-1}}$$

b) Covariance of Y_t at lag 2

$$Cov(Y_t, Y_{t-2}) = E[(0.8Y_{t-1} + e_t + 0.7e_{t-1} + 0.6e_{t-2})Y_{t-2}]$$

We are keeping terms that involve non-zero expectations. $= E[(0.8\,Y_{t-1} + 0.6e_{t-2})\,Y_{t-2}]$

i, $0.8\,Cov(Y_{t-1}, Y_{t-2}) = 0.8\gamma_1$.

ii, $0.6\,E[e_{t-2}Y_{t-2}]$, since e_{t-2} is independent of any noise terms at other times this becomes $0.6\,\sigma_e^2$.

$$\therefore Cov(Y_t, Y_{t-2}) = 0.8\gamma_1 + 0.6E(e_t Y_t)$$

$$= 0.8\gamma_1 + 0.6\,\sigma_e^2$$

$$\Rightarrow \boxed{e_2 = 0.8e_1 + \frac{0.6\sigma_e^2}{\gamma_0}}$$

4.13 sol) We are working with variances of 2 different sums of the random variables $Y_1, Y_2, Y_3, \dots Y_{n+1}$ where each Y_i has variance γ_0 and autocorrelation e_1 at lag 1.

1) Variance of the sum:-

$$Var(Y_{n+1}, Y_n + Y_{n-1} + \dots + Y_1) = (n+1) + 2ne_1)\gamma_0$$

$$= (1 + n(1 + 2e_1))\gamma_0$$

2) Variance of Alternating sum:-

$$Var(Y_{n+1} - Y_n + Y_{n-1} - \dots - Y_1) = ((n+1) - 2ne_1)\gamma_0$$

$$= (1 + n(1 - 2e_1))\gamma_0.$$

Conditions for variance non-negativity for both variances to be non-negative, we need:

$$\begin{cases} (1+n(1+2e_1)) \geq 0 \\ (1+n(1-2e_1)) \geq 0 \end{cases}$$

Expanding and simplifying we get

$$\Rightarrow \begin{cases} e_1 \geq -\frac{1}{2}\left(1+\frac{1}{n}\right) \\ e_1 \leq \frac{1}{2}\left(1+\frac{1}{n}\right) \end{cases}$$

i.e. $e_1 \geq -\frac{1}{2}$ and $e_1 \leq \frac{1}{2}$ for all n

The autocorrelation e_1 must satisfy $e_1 \leq \frac{1}{2}$ to ensure that the variances of both the sum and the alternating sum of Y_t remain non-negative.

4.16 sol). a) Given time series $Y_t = \phi Y_{t-1} + e_t$, $\phi = 3$

$$-\sum_{j=1}^{\infty}\left(\frac{1}{3}\right)^j e_{t+j} = 3\left(-\sum_{j=1}^{\infty}\left(\frac{1}{3}\right)^j e_{t-1+j}\right) + e_t$$

$$\Rightarrow -\sum_{j+1=2}^{\infty}\left(\frac{1}{3}\right)^{j+1} e_{t+j}.$$

b) $E(Y_t) = E\left(\sum_{j=1}^{\infty}\left(\frac{1}{3}\right)^j e_{t+j}\right) = 0.$

Since all terms are uncorrelated white noise.

$$Cov(Y_t, Y_{t-1}) = Cov\left(-\sum_{j=1}^{\infty}\left(\frac{1}{3}\right)^j e_{t+j}, \sum_{j=1}^{\infty}\left(\frac{1}{3}\right)^j e_{t+j-1}\right)$$

$$= Cov\left(-\frac{1}{3}e_{t+1}, -\frac{1}{3^2}e_{t+1}\right) + Cov\left(-\frac{1}{3}e_{t+2}, -\frac{1}{3^3}e_{t+2}\right) + \cdots + Cov\left(-\frac{1}{3^n}e_{t+n}, \frac{1}{3^{n+1}}e_{t+n}\right)$$

$$= \frac{1}{26}\sigma_e^2\left(1+\frac{1}{3}+\frac{1}{3^2}+\cdots+\frac{1}{3^n}\right)$$

$$\therefore Cov(Y_t, Y_{t-1}) = \frac{1}{26}\sigma_e^2\left(1+\frac{1}{3}+\frac{1}{3^2}+\cdots+\frac{1}{3^n}\right)$$

c) It is unsatisfactory because Y_t depends on future observations.

4.25 sol) a) $Y_t = \phi Y_{t-1} + e_t$

$$= \phi(\phi Y_{t-2} + e_{t-1}) + e_t$$

$$= \phi^2((\phi Y_{t-3} + e_{t-2}) + e_{t-1}) + e_t$$

$$\therefore Y_t = \phi^t Y_{t-1} + \phi e_{t-1} + \phi^2 e_{t-2} + \cdots + \phi^{t-1} e_1 + e_t.$$

b) $Y_t = \phi^t Y_0 + \phi e_{t-1} + \phi^2 e_{t-2} + \cdots + \phi^{t-1} e_1 + e_t$

$$E(Y_t) = E\left(\phi^t Y_0 + \sum_{j=0}^{t-1}\phi^j e_{t-j}\right)$$

$$E(Y_t) = \phi^t E(Y_0)$$

$$\boxed{\therefore E(Y_t) = \phi^t \mu_0}$$

c) $Y_t = \phi^t Y_0 + \sum_{k=0}^{t-1}\phi^k e_{t-k}$

$$Var(Y_t) = Var\left(\phi^t Y_0 + \sum_{k=0}^{t-1}\phi^k e_{t-k}\right)$$

$$= \phi^{2t}\sigma_0^2 + Var\left(\sum_{k=0}^{t-1}\phi^k e_{t-k}\right)$$

$$Var\left(\sum_{k=0}^{t-1}\phi^k e_{t-k}\right) = \sigma_e^2 \underbrace{\sum_{k=0}^{t-1}(\phi^2)^k}_{\frac{1-\phi^{2t}}{1-\phi^2}}$$

$$Var(Y_t) = \phi^{2t}\sigma_0^2 + \sigma_e^2\frac{1-\phi^{2t}}{1-\phi^2}$$

$$\boxed{\therefore Var(Y_t) = \sigma_0^2 + \sigma_e^2 t}$$

d) $Var(Y_t) = \phi^{2t}\sigma_0^2 + \sigma_e^2 \frac{1-\phi^{2t}}{1-\phi^2}$

$Var(Y_t) \approx \frac{\sigma_e^2}{1-\phi^2} \quad \{t \to \infty$

If $|\phi| > 1$ the $var(Y_t)$ either grows with t (if $|\phi| > 1$) or does not converge to a finite value (if $\phi = 1$) making the process not stationary.

If $\mu_0 = 0$ then $E(Y_t) = 0$ but for $Var(Y_t)$ to be free of t, ϕ cannot be 1.

e) $Var(Y_t) = \phi^2 \, Var(Y_{t-1}) + \sigma_e^2$

In stationary process $Var(Y_t)$ is same for all t
$Var(Y) = \phi^2 \, Var(X) + \sigma_e^2$

$Var(Y)(1-\phi^2) = \sigma_e^2$.

$\therefore Var(Y) = \frac{\sigma_e^2}{1-\phi^2}$

For the above equation to be finite and positive we require $1-\phi^2 > 0 \implies |\phi| < 1$

$\therefore |\phi| < 1$ is necessary to ensure that $Var(Y_t)$ remains finite and process is stationary.

Question 4:-

a) Autocorrelation for lag 1 for an MA(1) time series an MA(1) model can be represented as
$X_t = Y_t + \theta Y_{t-1}$

Where Y_t are white noise error terms with mean 0 and variance σ^2, θ is the parameter of the model to MA(1) model is defined as.

1) for Lag 0 (ACF(0)):
$e(0) = 1$

2) for Lag 1 (ACF(1))
$e(1) = \frac{-\theta}{1+\theta^2}$

3) for lags greater than 1 ACF(K) for $K > 1$
$e(K) = 0$

Given the parameter $\theta = 0.7$, we can calculate
$e(1) = \frac{0.7}{1+(0.7)^2} = \frac{0.7}{1.49}$

$e(1) \approx = 0.4698$

Autocorrelation at Lag 1 for the given MA(1) is approximately 0.47.

b) An AR(1) model can be represented as:

$$X_t = \phi X_{t-1} + e_t$$

For Lag k $e(k) = \phi^k = \frac{r_k}{r_0}$

Given parameter $\phi = 0.8$, we can now calculate $e(2)$ & $e(6)$

for Lag 2 $e(2) = \phi^2 = (0.8)^2 = 0.64$

or Lag 6, $e(6) = \phi^6 = (0.8)^6 = 0.262144$.